

# DIFFERENTIAL GEOMETRY AND LIE GROUPS

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Fifth Semester

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## Chapter 1

# PRELIMINARIES AND MANIFOLDS

July 20th.

The primary reference for this course will be Loring Tu's *An Introduction to Manifolds*. Prerequisites include Analysis of Several Variables, Linear Algebra, and Topology. The mid-term will account for 35% of the final grade, quizzes and student lectures will account for 15%, and the final exam will account for 50%.

Simply put, a manifold is any object that is locally Euclidean.

### 1.1 Calculus on $\mathbb{R}^n$

A point  $p$  or element in  $\mathbb{R}^n$  is associated with an  $n$ -tuple of coordinates  $x^1, x^2, \dots, x^n$ . Let  $U$  be an open neighbourhood of  $p$  in  $\mathbb{R}^n$ . For a non-negative integer  $k$ , we say that a function  $f : U \rightarrow \mathbb{R}$  is  $C^k$ -continuous at  $p \in U$  if all partial derivatives of  $f$  of order  $\leq k$  exist and are continuous at  $p$ .  $f$  is said to be  $C^\infty$  or *smooth* if  $f$  is  $C^k$  for all  $k \geq 0$ .  $f$  is  $C^k$  (or  $C^\infty$ ) on  $U$  if it is  $C^k$  (or  $C^\infty$ ) at every point in  $U$ .

If  $f$  happens to be a vector-valued function, i.e.  $f : U \rightarrow \mathbb{R}^m$ , we say  $f = (f^1, f^2, \dots, f^m)$  is  $C^k$  (or  $C^\infty$ ) at  $p$  if each component function  $f^i : U \rightarrow \mathbb{R}$  is  $C^k$  (or  $C^\infty$ ) at  $p$ .  $f$  is  $C^k$  (or  $C^\infty$ ) on  $U$  if it is  $C^k$  (or  $C^\infty$ ) at every point in  $U$ .

**Definition 1.1.** Let  $R$  be a commutative ring with unity. We term  $A$  to be an  $R$ -algebra if  $A$  is a commutative ring with unity and an  $R$ -module, and for  $r \in R$  and  $a, b \in A$ , we have  $r(ab) = (ra)b = a(rb)$ .

In this course, we will be primarily be concerned with the case when  $R$  is a field, and  $A$  is an  $R$ -vector space.

For a subset  $U \subseteq \mathbb{R}^n$ , we denote  $C^k(U)$  (or  $C^\infty(U)$ ) as the set of all  $C^k$  (or  $C^\infty$ ) functions from  $U$  to  $\mathbb{R}$ . Note that this is a commutative ring with unity. It is also a vector space over  $\mathbb{R}$ . Together,  $C^k(U)$  (or  $C^\infty(U)$ ) is a (commutative) algebra over  $\mathbb{R}$ . Moreover, we have the descending chain of subalgebras

$$C^0(U) \supseteq C^1(U) \supseteq C^2(U) \supseteq \dots C^k(U) \supseteq C^{k+1}(U) \supseteq \dots \quad (1.1)$$

with

$$\bigcap_{k \geq 0} C^k(U) = C^\infty(U). \quad (1.2)$$

$f : U \rightarrow \mathbb{R}$  is termed *real analytic* at  $p \in U$  if there exists a neighbourhood  $V \subseteq U$  of  $p$  such that

$$f(x) = f(p) + \sum_{i=1}^n \frac{\partial f}{\partial x^i}(p)(x^i - p^i) + \dots + \frac{1}{k!} \sum_{1 \leq i_1, \dots, i_k \leq n} \frac{\partial^k f}{\partial x^{i_1} \dots \partial x^{i_k}}(p)(x^{i_1} - p^{i_1}) \dots (x^{i_k} - p^{i_k}) + \dots \quad (1.3)$$

for all  $x \in V$ .  $f$  is said to be real analytic on  $U$  if it is real analytic at every point in  $U$ . We denote the set of all real analytic functions on  $U$  as  $C^\omega(U)$ . Note that  $C^\omega(U) \subseteq C^\infty(U)$ .

**Theorem 1.2** (*Taylor's theorem*). *Let  $p \in U \subseteq \mathbb{R}^n$ , and  $f : U \rightarrow \mathbb{R}$  be a  $C^\infty$  function. Further assume that for all  $x \in U$ , the line segment from  $p$  to  $x$  is contained in  $U$  ( $U$  in this case is said to be star-shaped with respect to  $p$ ). Then there exist  $g_1, \dots, g_n \in C^\infty(U)$  such that, for all  $x \in U$ ,*

$$f(x) = f(p) + \sum_{i=1}^n g_i(x)(x^i - p^i), \quad (1.4)$$

with

$$g_i(p) = \frac{\partial f}{\partial x^i}(p), \quad 1 \leq i \leq n. \quad (1.5)$$

*Proof.* Let  $x \in U$ . Then the line segment  $\{p + t(x - p) \mid t \in [0, 1]\}$  is contained in  $U$ . Composing  $f$  with the map  $t \mapsto p + t(x - p)$ , we obtain

$$\frac{df}{dt}(p + t(x - p)) = \sum_{i=1}^n \frac{\partial f}{\partial x^i}(p + t(x - p))(x^i - p^i). \quad (1.6)$$

Integrating from 0 to 1, we get

$$f(x) - f(p) = \int_0^1 \frac{df}{dt}(p + t(x - p))dt = \sum_{i=1}^n (x^i - p^i) \int_0^1 \frac{\partial f}{\partial x^i}(p + t(x - p))dt = \sum_{i=1}^n (x^i - p^i)g_i(x). \quad (1.7)$$

Clearly, because  $f$  is  $C^\infty$ , each  $g_i$  is also  $C^\infty$ . Further, we have  $g_i(p) = \int_0^1 \frac{\partial f}{\partial x^i}(p)dt = \frac{\partial f}{\partial x^i}(p)$ . ■

Let  $p = (p^1, \dots, p^n)$ ,  $v \in \mathbb{R}^n$ , and  $f : U \rightarrow \mathbb{R}$  be  $C^\infty$  at  $p$ . The *directional derivative* of  $f$  at  $p$  in the direction of  $v$  is defined as

$$\lim_{t \rightarrow 0} \frac{f(p + tv) - f(p)}{t} = \left. \frac{d}{dt} f(p + tv) \right|_{t=0} = \sum_{i=1}^n \frac{df}{dx^i}(p)v^i. \quad (1.8)$$

From here, we obtain the operator

$$D_v = \sum_{i=1}^n v^i \frac{\partial}{\partial x^i} \Big|_p \quad (1.9)$$

Now consider all pairs of the form  $(f, U)$  where  $p \in U \subseteq \mathbb{R}^n$  is an open neighbourhood of  $p$  and  $f : U \rightarrow \mathbb{R}$  is  $C^\infty$ . Define an equivalence relation  $\sim$  on this set by declaring  $(f, U) \sim (g, V)$  if and only if there exists an open neighbourhood  $W \subseteq U \cap V$  of  $p$  such that  $f|_W = g|_W$ . The equivalence class of  $(f, U)$  is denoted by  $[(f, U)]$ , and the set of all such equivalence classes is denoted by  $C_p^\infty(\mathbb{R}^n)$ . Such an equivalence class  $[(f, U)]$  is called the *germ* of  $f$  at  $p$ . We can give  $C_p^\infty(\mathbb{R}^n)$  the structure of a commutative ring with unity by defining

$$[(f, U)] + [(g, V)] = [(f + g, U \cap V)], \quad [(f, U)] \cdot [(g, V)] = [(fg, U \cap V)], \quad 1 = [(1, \mathbb{R}^n)]. \quad (1.10)$$

It can also be given the structure of a vector space over  $\mathbb{R}$  by defining

$$\lambda[(f, U)] = [(\lambda f, U)], \quad \lambda \in \mathbb{R}. \quad (1.11)$$

Thus,  $C_p^\infty(\mathbb{R}^n)$  is an  $\mathbb{R}$ -algebra. Now, given a germ  $\alpha = [(f, U)] \in C_p^\infty(\mathbb{R}^n)$  and a vector  $v \in \mathbb{R}^n$ , we define

$$D_v \alpha = \sum_{i=1}^n v^i \frac{\partial f}{\partial x^i} \Big|_p = v \cdot \nabla f(p). \quad (1.12)$$

Note that the choice of  $f$  is immaterial, since if  $[(f, U)] = [(g, V)]$ , then there exists an open neighbourhood  $W \subseteq U \cap V$  of  $p$  such that  $f|_W = g|_W$ , and hence  $\nabla f(p) = \nabla g(p)$ . One can infer that  $D_{\lambda v}(f) = \lambda D_v(f) =$

$D_v(\lambda f)$  and  $D_v(f + g) = D_v(f) + D_v(g)$  for all  $\lambda \in \mathbb{R}$  and  $f, g \in C_p^\infty(\mathbb{R}^n)$ . Thus,  $D_v$  is an  $\mathbb{R}$ -linear map from  $C_p^\infty(\mathbb{R}^n)$  to  $\mathbb{R}$  (hereforth, we shall use  $f$  to also denote the germ  $[(f, U)]$  of  $f$  at  $p$ ). Also,

$$D_v(fg) = \sum_{i=1}^n v^i \frac{\partial(fg)}{\partial x^i}(p) = \sum_{i=1}^n v^i \left( f(p) \frac{\partial g}{\partial x^i}(p) + g(p) \frac{\partial f}{\partial x^i}(p) \right) = f(p)D_v(g) + g(p)D_v(f). \quad (1.13)$$

Such a property makes  $D_v$  what we call a derivation.

**Definition 1.3.** Any  $\mathbb{R}$ -linear map  $T : C_p^\infty(\mathbb{R}^n) \rightarrow \mathbb{R}$  satisfying the Leibniz rule  $T(fg) = f(p)T(g) + g(p)T(f)$  for all  $f, g \in C_p^\infty(\mathbb{R}^n)$  is called a *derivation* at  $p$ .

One may show that if  $T$  is a derivation at  $p$ , then  $T(1) = 0$ .

The set of all derivations at  $p$  is denoted by  $\mathfrak{D}_p(\mathbb{R}^n)$ . It is easy to see that  $\mathfrak{D}_p(\mathbb{R}^n)$  is a vector space over  $\mathbb{R}$ . Now consider the map

$$\theta : \mathbb{R}^n \rightarrow \mathfrak{D}_p(\mathbb{R}^n), \quad v \mapsto D_v. \quad (1.14)$$

This  $\theta$  is an  $\mathbb{R}$ -linear map (verify). We claim that  $\theta$  is an isomorphism of real vector spaces. To this end, let  $v \in \mathbb{R}^n$  such that  $D_v = 0 \in \mathfrak{D}_p(\mathbb{R}^n)$ . Then  $D_v f = 0$  for all  $f \in C_p^\infty(\mathbb{R}^n)$ . In particular, for the coordinate functions  $x^i$ , we have  $D_v x^i = v_i = 0$  for all  $1 \leq i \leq n$ . Thus,  $v = 0$ , and hence  $\theta$  is injective. To show surjectivity, we appeal to Taylor's theorem; let  $D \in \mathfrak{D}_p(\mathbb{R}^n)$ . We claim that  $D = D_v$  where  $v = (D(x^1), \dots, D(x^n))$ . Indeed, for any  $f \in C_p^\infty(\mathbb{R}^n)$ , we have

$$\begin{aligned} D(f) &= D(f(p)) + D\left(\sum_{i=1}^n (x^i - p^i)g_i(x)\right) = 0 + \sum_{i=1}^n ((x^i - p^i)(p)Dg_i(x) + D(x^i - p^i)g_i(p)) \\ &= \sum_{i=1}^n (Dx^i)g_i(p) = \sum_{i=1}^n \frac{\partial f}{\partial x^i}(p)Dx^i = D_v(f). \end{aligned} \quad (1.15)$$

Hence,  $\theta$  is surjective, and we have the desired isomorphism  $\mathfrak{D}_p(\mathbb{R}^n) \cong \mathbb{R}^n$ .

Note that a germ  $f \in C_p^\infty(\mathbb{R}^n)$  has a valid value at  $p$ , but no other point. Hence, we have the evaluation map

$$\text{ev}_p : C_p^\infty(\mathbb{R}^n) \rightarrow \mathbb{R}, \quad [(f, U)] \mapsto f(p). \quad (1.16)$$

The kernel here is  $\{[(f, U)] \in C_p^\infty(\mathbb{R}^n) \mid f(p) = 0\}$ , which is an ideal in  $C_p^\infty(\mathbb{R}^n)$ . For any other element  $[(f, U)] \in C_p^\infty(\mathbb{R}^n)$  with  $f(p) \neq 0$ , a valid inverse is given by  $[(f^{-1}, U \setminus \{f = 0\})]$ . This makes  $C_p^\infty(\mathbb{R}^n)$  a local ring.

## 1.2 Multilinear Algebra

*July 22nd.*

Hereforth,  $V$  shall denote a finite-dimensional real vector space, and  $V^V$  shall denote its *dual space*  $\text{Hom}_{\mathbb{R}}(V, \mathbb{R})$ . If  $e_1, \dots, e_n$  are a basis for  $V$ , then the *dual basis* of  $V^V$  is denoted by  $\alpha^1, \dots, \alpha^n \in V^V$ , and is defined by  $\alpha^i(e_j) = \delta_{i,j}$  for all  $1 \leq i, j \leq n$ . One may verify that the dual basis  $\{\alpha^1, \dots, \alpha^n\}$ , induced by the basis  $\{e_1, \dots, e_n\}$ , is indeed a basis for  $V^V$ .  $S_k$  shall denote the permutation group on  $k$  symbols, satisfying  $|S_k| = k!$ . For any  $\sigma \in S_k$ , the sign of  $\sigma$  is denoted by  $\text{sgn}(\sigma)$ , and is defined as  $+1$  if  $\sigma$  is an even permutation, and  $-1$  if  $\sigma$  is an odd permutation.

With notation out of the way, let  $k \geq 1$  be an integer. We shall consider the  $k$ -tuple  $V^k = V \times \dots \times V$  ( $k$  times).

**Definition 1.4.** A mapping  $f : V^k \rightarrow \mathbb{R}$  is said to be *k-linear* if  $f$  is linear in each of its  $k$  arguments.

A special case is when  $k = 2$ , in which case  $f$  is called a *bilinear form*. Such an  $f : V \times V \rightarrow \mathbb{R}$  satisfies

$$f(\lambda v + \mu w, u) = \lambda f(v, u) + \mu f(w, u), \quad f(u, \lambda v + \mu w) = \lambda f(u, v) + \mu f(u, w). \quad (1.17)$$

A common example of a bilinear form is any valid inner product  $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$  on  $V$ . Another example is the determinant  $\det : \mathbb{R}^n \rightarrow \mathbb{R}$ , where the vectors are interpreted as the columns of a matrix. We can also write

$$L_k(V) = \{f : V^k \rightarrow \mathbb{R} \mid f \text{ is } k\text{-linear}\}. \quad (1.18)$$

It is easy to see that  $L_k(V)$  is an  $\mathbb{R}$ -vector space.

**Definition 1.5.** A  $k$ -linear map  $f : V^k \rightarrow \mathbb{R}$  is called a *symmetric  $k$ -linear form* if for all  $\sigma \in S_k$ ,

$$f(v_1, \dots, v_k) = f(v_{\sigma(1)}, \dots, v_{\sigma(k)}). \quad (1.19)$$

That is,  $f$  is invariant under any permutation of its arguments. Moreover,  $f$  is called an *alternating  $k$ -linear form* if for all  $\sigma \in S_k$ ,

$$f(v_1, \dots, v_k) = \text{sgn}(\sigma) f(v_{\sigma(1)}, \dots, v_{\sigma(k)}). \quad (1.20)$$

One can see that the standard inner product on  $\mathbb{R}^n$  is a symmetric bilinear form, and the determinant is an alternating  $n$ -linear form. Related to these definitions, we can immediately define the following subspaces of  $L_k(V)$ :

$$S_k(V) = \{f \in L_k(V) \mid f \text{ is symmetric}\} \subseteq L_k(V), \quad A_k(V) = \{f \in L_k(V) \mid f \text{ is alternating}\} \subseteq L_k(V). \quad (1.21)$$

Currently, our main interest of study is  $A_k(V)$ , the space of alternating  $k$ -linear forms on  $V$ . Elements of  $A_k(V)$  are termed *alternating  $k$ -tensors*. Similarly, those of  $S_k(V)$  and  $L_k(V)$  are called *symmetric  $k$ -tensors* and  *$k$ -tensors*, respectively. For  $k = 0$ , we define  $L_0(V) = S_0(V) = A_0(V) = \mathbb{R}$ . For  $k = 1$ , we have  $L_1(V) = S_1(V) = A_1(V) = V^V$ . We can define the (left) action of  $S_k$  on  $L_k(V)$  by

$$(\sigma \cdot f)(v_1, \dots, v_k) = f(v_{\sigma(1)}, \dots, v_{\sigma(k)}). \quad (1.22)$$

Thus,  $f$  is symmetric if  $\sigma \cdot f = f$  for all  $\sigma \in S_k$ , and  $f$  is alternating if  $\sigma \cdot f = \text{sgn}(\sigma)f$  for all  $\sigma \in S_k$ .

Given an  $f \in L_k(V)$ , we can define the *symmetrization*  $Sf \in S_k(V)$ , of  $f$ , as

$$Sf = \sum_{\sigma \in S_k} \sigma \cdot f. \quad (1.23)$$

One may, indeed, check that  $Sf$  is symmetric. Similarly, we can define the *alternation*  $Af \in A_k(V)$ , of  $f$ , as

$$Af = \sum_{\sigma \in S_k} \text{sgn}(\sigma) \sigma \cdot f. \quad (1.24)$$

### 1.2.1 The Tensor and Wedge Products

July 24th.

For any  $f \in L_k(V)$  and  $g \in L_l(V)$ , we can define the *tensor product*  $f \otimes g \in L_{k+l}(V)$  as

$$(f \otimes g)(v_1, \dots, v_{k+l}) = f(v_1, \dots, v_k)g(v_{k+1}, \dots, v_{k+l}). \quad (1.25)$$

Verify that  $f \otimes g$  is indeed  $(k+l)$ -linear. We are interested in the tensor product of alternating tensors, so let  $f \in A_k(V)$  and  $g \in A_l(V)$ . Then, we can define the *wedge product*  $f \wedge g \in A_{k+l}(V)$  as

$$f \wedge g = \frac{1}{k!l!} A(f \otimes g) = \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \sigma \cdot (f \otimes g). \quad (1.26)$$

The associativity of the tensor product holds easily: for  $f \in L_k(V)$ ,  $g \in L_l(V)$ , and  $h \in L_m(V)$ ,

$$(f \otimes g) \otimes h = f \otimes (g \otimes h). \quad (1.27)$$

**Proposition 1.6.** For  $f \in A_k(V)$  and  $g \in A_l(V)$ , we have

$$f \wedge g = (-1)^{kl} g \wedge f. \quad (1.28)$$

*Proof.* We shall show that  $A(f \otimes g) = (-1)^{kl} A(g \otimes f)$ . To this end, fix the permutation  $\tau \in S_{k+l}$  defined by

$$\tau = \begin{pmatrix} 1 & 2 & \cdots & l & l+1 & l+2 & \cdots & l+k \\ k+1 & k+2 & \cdots & k+l & 1 & 2 & \cdots & k \end{pmatrix}. \quad (1.29)$$

We have

$$\begin{aligned} A(f \otimes g)(v_1, \dots, v_{k+l}) &= \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) f(v_{\sigma(1)}, \dots, v_{\sigma(k)}) g(v_{\sigma(k+1)}, \dots, v_{\sigma(k+l)}) \\ &= \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) f(v_{\sigma\tau(l+1)}, \dots, v_{\sigma\tau(l+k)}) g(v_{\sigma\tau(1)}, \dots, v_{\sigma\tau(l)}) \\ &= \text{sgn}(\tau) \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma\tau) f(v_{\sigma\tau(l+1)}, \dots, v_{\sigma\tau(l+k)}) g(v_{\sigma\tau(1)}, \dots, v_{\sigma\tau(l)}) \\ &= \text{sgn}(\tau) A(g \otimes f)(v_1, \dots, v_{k+l}). \end{aligned} \quad (1.30)$$

With  $\text{sgn}(\tau) = (-1)^{kl}$ , the proof is complete.  $\blacksquare$

**Corollary 1.7.** For  $f \in A_k(V)$ , we have  $f \wedge f = 0$  if  $k$  is odd.

*Proof.* Easy since  $f \wedge f = (-1)^{k^2} f \wedge f = -f \wedge f$  if  $k$  is odd, so  $f \wedge f = 0$ .  $\blacksquare$

**Lemma 1.8.** For  $f \in L_k(V)$  and  $g \in L_l(V)$ , we have

$$A(Af \otimes g) = k! A(f \otimes g), \quad A(f \otimes Ag) = l! A(f \otimes g). \quad (1.31)$$

*Proof.* We shall prove the first equality; the second follows similarly. We have

$$\begin{aligned} A(Af \otimes g) &= \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \sigma \cdot \left( \sum_{\tau \in S_k} \text{sgn}(\tau) (\tau \cdot f) \otimes g \right) = \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \sigma \cdot \left( \sum_{\tau \in S_k} \text{sgn}(\tau) \tau \cdot (f \otimes g) \right) \\ &= \sum_{\tau \in S_k} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma\tau) \sigma\tau \cdot (f \otimes g) = k! \sum_{\alpha \in S_{k+l}} \text{sgn}(\alpha) \alpha \cdot (f \otimes g) = k! A(f \otimes g). \end{aligned} \quad (1.32)$$

**Proposition 1.9.** The wedge product is associative, i.e. for  $f \in A_k(V)$ ,  $g \in A_l(V)$ , and  $h \in A_m(V)$ , we have

$$(f \wedge g) \wedge h = f \wedge (g \wedge h). \quad (1.33)$$

*Proof.* We have

$$\begin{aligned} (f \wedge g) \wedge h &= \frac{1}{(k+l)!m!} A((f \wedge g) \otimes h) = \frac{1}{(k+l)!m!k!l!} A(A(f \otimes g) \otimes h) = \frac{(k+l)!}{(k+l)!l!k!m!} A((f \otimes g) \otimes h) \\ &= \frac{1}{k!l!m!} A(f \otimes g \otimes h) = f \wedge (g \wedge h). \end{aligned} \quad (1.34)$$

**Corollary 1.10.**  $f \wedge g \wedge h = \frac{1}{k!l!m!} A(f \otimes g \otimes h)$  for  $f \in A_k(V)$ ,  $g \in A_l(V)$ , and  $h \in A_m(V)$ . Inductively,

$$f_1 \wedge \cdots \wedge f_k = \frac{1}{k_1! \cdots k_k!} A(f_1 \otimes \cdots \otimes f_k), \quad (1.35)$$

where  $f_i \in A_{k_i}(V)$  for  $1 \leq i \leq k$ .

**Proposition 1.11.** Let  $\alpha^1, \dots, \alpha^n \in V^V = A_1(V) = L_1(V)$ , and  $v_1, \dots, v_k \in V$ . Then

$$(\alpha^1 \wedge \cdots \wedge \alpha^k)(v_1, \dots, v_k) = \det(\alpha^i(v_j))_{1 \leq i, j \leq k}. \quad (1.36)$$

*Proof.* We have

$$\begin{aligned} (\alpha^1 \wedge \cdots \wedge \alpha^k)(v_1, \dots, v_k) &= \sum_{\sigma \in S_k} \text{sgn}(\sigma) (\alpha^1 \otimes \cdots \otimes \alpha^k)(v_{\sigma(1)}, \dots, v_{\sigma(k)}) \\ &= \sum_{\sigma \in S_k} \text{sgn}(\sigma) \alpha^1(v_{\sigma(1)}) \alpha^2(v_{\sigma(2)}) \cdots \alpha^k(v_{\sigma(k)}) = \det(\alpha^i(v_j))_{1 \leq i, j \leq k}. \end{aligned} \quad (1.37)$$

■

*July 27th.*

Let  $V$  be an  $\mathbb{R}$ -vector space with basis  $e_1, \dots, e_n$ , and  $V^V$  be its dual space with dual basis  $\alpha^1, \dots, \alpha^n$ . Moreover, let  $I$  be a multi-index of length  $k$ , i.e.  $I = (i_1, \dots, i_k)$  with  $1 \leq i_1, \dots, i_k \leq n$ . We define  $e_I := (e_{i_1}, \dots, e_{i_k}) \in V^k$  and  $\alpha^I := \alpha^{i_1} \wedge \cdots \wedge \alpha^{i_k} \in A_k(V)$ . Then, a  $k$ -linear form  $f \in L_k(V)$  is completely determined by its values on the  $e_I$ , and an alternating  $k$ -linear form  $f \in A_k(V)$  is completely determined by its values on the  $e_I$  with  $i_1 < i_2 < \cdots < i_k$ .

To this end, we first claim that  $\alpha^I(e_J) = \delta_J^I$  for all multi-indices  $I, J$  of length  $k$ . Indeed, if  $J = (j_1, \dots, j_k)$  with  $j_1 < j_2 < \cdots < j_k$ , then

$$\alpha^I(e_J) = \det(\alpha^{i_m}(e_{j_n}))_{1 \leq m, n \leq k}; \quad (1.38)$$

. this is 1 for  $I = J$ . Suppose  $I \neq J$ . Let  $1 \leq l \leq k$  be the first index where  $i_l \neq j_l$ . Then the  $l^{\text{th}}$  row of  $\alpha^{i_m}(e_{j_n})_{1 \leq m, n \leq k}$  is an all zeroes row, and hence  $\det(\alpha^{i_m}(e_{j_n}))_{1 \leq m, n \leq k} = 0$ . Thus,  $\alpha^I(e_J) = \delta_J^I$ . We now claim that  $\alpha^I$ , for strictly increasing multi-indices  $I$  of length  $k$ , form a basis for  $A_k(V)$ . For linear independence, we simply have (the sums in this context are over strictly increasing multi-indices  $I$  of length  $k$ ):

$$\sum_I c_I \alpha^I = 0 \implies \sum_I c_I \alpha^I(e_J) = 0 \implies c_J = 0, \quad \forall J. \quad (1.39)$$

To show that they span  $A_k(V)$ , let  $f \in A_k(V)$ . Then consider the  $k$ -linear form

$$g = \sum_I f(e_I) \alpha^I. \quad (1.40)$$

We show that  $f = g$ ; it is enough to prove that  $f(e_J) = g(e_J)$  for all strictly increasing multi-indices  $J$  of length  $k$ . We have

$$g(e_J) = \sum_I f(e_I) \alpha^I(e_J) = f(e_J) \alpha^J(e_J) + \sum_{I \neq J} f(e_I) \alpha^I(e_J) = f(e_J). \quad (1.41)$$

**Corollary 1.12.**  $A_k(V) = 0$  if  $k > n$ . Further,  $\dim A_k(V) = \binom{n}{k}$  if  $1 \leq k \leq n$ .

### 1.3 Building Blocks for Manifolds

Recall that we had defined the class of germs  $C_p^\infty(\mathbb{R}^n)$  of  $C^\infty$  functions at a point  $p \in \mathbb{R}^n$ , which agree on some neighbourhood of  $p$ . One can show that  $C_p^\infty(\mathbb{R}^n)$  and  $C_p^\infty(U)$  are isomorphic as  $\mathbb{R}$ -algebras, where  $U \subseteq \mathbb{R}^n$  is an open neighbourhood of  $p$ . Note that we term  $T_p(U) = T_p(\mathbb{R}^n) = \mathbb{R}^n$  as the *tangent space* of  $U$  (or  $\mathbb{R}$ ) at  $p$ .

**Definition 1.13.** Let  $U \subseteq \mathbb{R}^n$  be an open subset. A *vector field* on  $U$  is a function  $X : U \rightarrow \coprod_{p \in U} T_p(\mathbb{R}^n)$  such that  $X(p) \in T_p(\mathbb{R}^n)$  for all  $p \in U$ . In particular,

$$X(p) = \sum_{i=1}^n a^i(p) \frac{\partial}{\partial x^i} \Big|_p, \quad a^i(p) \in \mathbb{R}. \quad (1.42)$$

We can rewrite this as

$$X = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}, \quad a^i : U \rightarrow \mathbb{R}. \quad (1.43)$$

$X$  is called a *smooth vector field* if each  $a^i$  is a smooth function on  $U$ . The set of all smooth vector fields on  $U$  is denoted by  $\mathfrak{X}(U)$ .

This is a real vector space; note that for  $X, Y \in \mathfrak{X}(U)$ , we have  $X + Y \in \mathfrak{X}(U)$ , and for  $\lambda \in \mathbb{R}$ , we have  $\lambda X \in \mathfrak{X}(U)$ . Further,  $\mathfrak{X}(U)$  is a  $C^\infty(U)$ -module since for  $f \in C^\infty(U)$  and  $X \in \mathfrak{X}(U)$ , we have  $fX \in \mathfrak{X}(U)$ , where

$$(fX)(p) = f(p)X(p), \quad p \in U. \quad (1.44)$$

Notice that  $X \in \mathfrak{X}(U)$  describes a map  $C^\infty(U) \rightarrow C^\infty(U)$ , where  $f \mapsto Xf$ , being defined as  $(Xf)(p) = X_p f$ . This is a derivation on  $C^\infty(U)$ , since for  $f, g \in C^\infty(U)$ , we have

$$(X(fg))(p) = X_p(fg) = f(p)X_p g + g(p)X_p f = (fXg)(p) + (gXf)(p). \quad (1.45)$$

We refine our definition of a derivation as follows.

**Definition 1.14.** Let  $K$  be a field, and  $A$  be a  $K$ -algebra. A *derivation*  $D : A \rightarrow A$  is an  $\mathbb{R}$ -linear map satisfying the Leibniz rule  $D(ab) = aD(b) + bD(a)$  for all  $a, b \in A$ .

**Definition 1.15.** A *differential 1-form* on  $U$  is a smooth function  $\omega : U \rightarrow \coprod_{p \in U} T_p(\mathbb{R}^n)^V$  such that  $\omega(p) \in T_p(\mathbb{R}^n)^V =: T_p^*(\mathbb{R}^n)$  for all  $p \in U$ .

For example, for  $f \in C^\infty(U)$ , we can define the differential 1-form  $df$  on  $U$  by  $df(p) = df_p \in \text{Hom}_{\mathbb{R}}(T_p \mathbb{R}^n, \mathbb{R})$ , where  $df_p(X) = X_p f$  for all  $X \in T_p \mathbb{R}^n$ . In particular,  $dx^i$  is a differential 1-form on  $U$  for all  $1 \leq i \leq n$ . In fact,  $\{dx^1, \dots, dx^n\}$  is the dual basis of  $T_p^* \mathbb{R}^n$  corresponding to the basis  $\{\frac{\partial}{\partial x^1} \Big|_p, \dots, \frac{\partial}{\partial x^n} \Big|_p\}$  of  $T_p \mathbb{R}^n$ . Thus, a differential 1-form  $\omega$  on  $U$  can be expressed as

$$\omega(p) = \sum_{i=1}^n a_i(p) dx^i \Big|_p, \quad a_i : U \rightarrow \mathbb{R}. \quad (1.46)$$

$T_p^*(\mathbb{R}^n)$ , here, is termed the *cotangent space* of  $\mathbb{R}^n$  at  $p$ . Thus, our current setup is as follows:  $V = T_p \mathbb{R}^n$ , and  $V^V = T_p^* \mathbb{R}^n$ . The basis for  $T_p \mathbb{R}^n$  considered is  $\{\frac{\partial}{\partial x^1} \Big|_p, \dots, \frac{\partial}{\partial x^n} \Big|_p\}$ , and the corresponding dual basis for  $T_p^* \mathbb{R}^n$  is  $\{dx^1 \Big|_p, \dots, dx^n \Big|_p\}$ . And finally, we have

$$A_k(T_p \mathbb{R}^n) = \text{the space of alternating } k\text{-linear forms on } T_p \mathbb{R}^n \quad (1.47)$$

which has the basis  $\{dx^{i_1} \Big|_p \wedge \dots \wedge dx^{i_k} \Big|_p \mid 1 \leq i_1 < \dots < i_k \leq n\}$ .

**Definition 1.16.** A *differential  $k$ -form* on  $U$  is a smooth function  $\omega : U \rightarrow \coprod_{p \in U} A_k(T_p \mathbb{R}^n)$  such that  $\omega(p) \in A_k(T_p \mathbb{R}^n)$  for all  $p \in U$ .

Any such  $\omega$  may be written as  $\omega = \sum_I a_I dx_I$ . It is termed a smooth differential  $k$ -form if each  $a_I$  is a smooth function on  $U$ . The set of all smooth differential 1-forms on  $U$  is denoted by  $\Omega^1(U)$ , and the set

of all smooth differential  $k$ -forms on  $U$  is denoted by  $\Omega^k(U)$ . Both these sets are  $\mathbb{R}$ -vector spaces, and  $\Omega^k(U)$  is a free  $C^\infty(U)$ -module of rank  $\binom{n}{k}$ . The isomorphism may be given as

$$\Omega^1(U) \cong C^\infty(U)^{\oplus \binom{n}{1}}, \quad \omega = \sum_I a_I dx_I \mapsto (a_1, a_2, \dots, a_{\binom{n}{1}}). \quad (1.48)$$

We can endow  $\Omega^k(U)$  with the wedge product  $\wedge : \Omega^k(U) \times \Omega^l(U) \rightarrow \Omega^{k+l}(U)$ , defined as

$$(\omega \wedge \eta)(p) = \omega(p) \wedge \eta(p), \quad p \in U, \quad (1.49)$$

or

$$\omega \wedge \eta = \sum_{I, J} a_I b_J dx_I \wedge dx_J \quad (1.50)$$

where  $I$  and  $J$  are strictly increasing multi-indices of length  $k$  and  $l$ , respectively. Note that if  $I \cap J \neq \emptyset$ , then  $dx_I \wedge dx_J = 0$ . Thus, the sum may be taken over all strictly increasing multi-indices  $I$  and  $J$  of length  $k$  and  $l$ , respectively, such that  $I \cap J = \emptyset$ . Also,

$$A_*(V) = \bigoplus_{k=0}^n A_k(V) \quad (1.51)$$

is defined as the *exterior algebra* of  $V$ . The wedge product on here is associative and anti-commutative. We define  $\Omega^0(U) = C^\infty(U)$ . The map  $d : \Omega^k(U) \rightarrow \Omega^{k+1}(U)$  is defined as the *exterior derivative*; for  $k = 0$ , we have

$$d : C^\infty(U) \rightarrow \Omega^1(U), \quad f \mapsto df = \sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i. \quad (1.52)$$

For a  $k$ -form  $\omega = \sum_I a_I dx_I \in \Omega^k(U)$ , we have

$$d\omega = \sum_I da_I \wedge dx_I = \sum_I \sum_{j=1}^n \frac{\partial a_I}{\partial x^j} dx^j \wedge dx_I. \quad (1.53)$$

**Example 1.17.** Let  $U \subseteq \mathbb{R}^3$ . Let us call the coordinates  $x^1 = x$ ,  $x^2 = y$ , and  $x^3 = z$ . The only non-zero forms are  $\Omega^0(U)$ ,  $\Omega^1(U)$ ,  $\Omega^2(U)$ , and  $\Omega^3(U)$ . Note that  $\Omega^0(U) = C^\infty(U)$ , and there's nothing to say about this. For  $\Omega^1(U)$ , a form looks like  $f dx + g dy + h dz$  for  $f, g, h \in C^\infty(U)$ . For  $\Omega^2(U)$ , a form looks like  $R dx \wedge dy + Q dx \wedge dz + P dy \wedge dz$  for  $f, g, h \in C^\infty(U)$ .

*July 29th.*

The map  $\Omega^1(U) \times \mathfrak{X}(U) \rightarrow C^\infty(U)$ , defined as  $(\omega, X) \mapsto \omega(X)$ , is  $\mathbb{R}$ -bilinear. In particular, for  $\omega = \sum_{i=1}^n a_i dx^i$  and  $X = \sum_{j=1}^n b^j \frac{\partial}{\partial x^j}$ , we have

$$\omega(X) = \sum_{i=1}^n a_i dx^i \left( \sum_{j=1}^n b^j \frac{\partial}{\partial x^j} \right) = \sum_{i=1}^n a_i b^i \in C^\infty(U). \quad (1.54)$$

Extending this, we have the map  $\Omega^k(U) \times \mathfrak{X}(U)^k \rightarrow C^\infty(U)$ , defined as  $(\omega, X_1, \dots, X_k) \mapsto \omega(X_1, \dots, X_k)$ , which is  $\mathbb{R}$ -multilinear. In particular, for  $\omega = \sum_I b_I dx^I$  and  $X_i = \sum_{j=1}^n a_i^j \frac{\partial}{\partial x^j}$ , we have

$$\omega(X_1, \dots, X_k) = \sum_I b_I a_{i_1}^1 \cdots a_{i_k}^k \in C^\infty(U) \quad (1.55)$$

using the fact that  $dx^I(\frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_k}}) = 1$  if the multi-index matches, and 0 otherwise.

### 1.3.1 Exterior Differentiation

Here,  $d_k$  shall denote the map  $d_k : \Omega^k(U) \rightarrow \Omega^{k+1}(U)$ , defined as the *exterior derivative*. The subscript  $k$  is often omitted, and we simply write  $d : \Omega^k(U) \rightarrow \Omega^{k+1}(U)$ . For  $k = 0$ , we have

$$d : C^\infty(U) \rightarrow \Omega^1(U), \quad f \mapsto df = \sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i. \quad (1.56)$$

Analogously, for a general  $k$ -form  $\omega = \sum_I a_I dx^I \in \Omega^k(U)$ , we have

$$d : \Omega^k(U) \rightarrow \Omega^{k+1}(U), \quad \omega \mapsto d\omega = \sum_I da_I \wedge dx^I = \sum_I \sum_{j=1}^n \frac{\partial a_I}{\partial x^j} dx^j \wedge dx^I. \quad (1.57)$$

All the  $d$  maps are  $\mathbb{R}$ -linear, but not  $C^\infty(U)$ -linear. Also, for two forms  $\omega \in \Omega^k(U)$  and  $\eta \in \Omega^l(U)$ , we have

$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta. \quad (1.58)$$

This is known as *antiderivation*. Moreover,  $d_{k+1} \circ d_k = d^2 = 0$ . Both these properties may be verified by the reader. Now suppose we have another map  $D : \Omega^k(U) \rightarrow \Omega^{k+1}(U)$  which is  $\mathbb{R}$ -linear, satisfies antiderivation, and  $D^2 = 0$ , and  $Df = df$  for all  $f \in C^\infty(U)$ . Then  $D = d$ ; the exterior derivative is unique. Thus, a chain of maps is formed:

$$0 \xrightarrow{d} C^\infty(U) \xrightarrow{d} \Omega^1(U) \xrightarrow{d} \Omega^2(U) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^n(U) \xrightarrow{d} 0 \quad \text{for open } U \subseteq \mathbb{R}^n. \quad (1.59)$$

An  $\omega \in \Omega^k(U)$  is called an *exact  $k$ -form* if  $\omega = d\eta$  for some  $\eta \in \Omega^{k-1}(U)$ . It is called a *closed  $k$ -form* if  $d\omega = 0$ . Note that every exact form is closed, but the converse is not true in general. We will refine the following later: quotienting the space of closed  $k$ -forms by the space of exact  $k$ -forms gives us the *de Rham cohomology* of  $U$ , denoted by  $H_{\text{dR}}^k(U)$ .

## 1.4 Manifolds

July 31st.

**Definition 1.18.** A topological space  $M$  is termed a *topological manifold* of dimension  $n$  if

1.  $M$  is Hausdorff,
2.  $M$  is second countable (that is, has a countable basis), and
3. for all  $p \in M$ , there exists an open neighbourhood  $U \subseteq M$  and a homeomorphism  $\varphi : U \rightarrow \varphi(U) \subseteq \mathbb{R}^n$ , where  $\varphi(U)$  is open. This makes a manifold *locally euclidean*.

The *invariance of dimension* property states that if we have two open sets  $U \subseteq \mathbb{R}^n$  and  $V \subseteq \mathbb{R}^m$  such that  $U$  is homeomorphic to  $V$ , then  $m = n$ . Thus, the dimension of a topological manifold is well-defined.

**Definition 1.19.** A pair  $(U, \varphi)$  is called a *coordinate chart* on  $M$  if  $U \subseteq M$  is open and  $\varphi : U \rightarrow \varphi(U) \subseteq \mathbb{R}^n$  is a homeomorphism.

A trivial example is  $\mathbb{R}^n$  with one (coordinate) chart  $(\mathbb{R}^n, \text{id}_{\mathbb{R}^n})$ . In fact, if  $U \subseteq \mathbb{R}^n$  is open, then  $(U, \text{id}_U)$  is a coordinate chart. Note that subspaces inherit the properties of Hausdorffness and second countability. We state some properties without proof; if  $X$  is locally Euclidean but not  $T_2$ , then  $X$  being a topological manifold implies that the number of connected (or path connected) components is at most countable.

**Definition 1.20.** Two charts  $(U, \varphi)$  and  $(V, \psi)$  on a topological manifold  $M$  are said to be  *$C^\infty$ -compatible* if either

1.  $U \cap V = \emptyset$ , or
2.  $U \cap V \neq \emptyset$ , and the transition maps  $\psi \circ \varphi^{-1} : \varphi(U \cap V) \rightarrow \psi(U \cap V)$  and  $\varphi \circ \psi^{-1} : \psi(U \cap V) \rightarrow \varphi(U \cap V)$  are  $C^\infty$ -maps between open subsets of  $\mathbb{R}^n$ .

Similarly,  $C^k$ -compatibility may be defined for  $k \geq 0$ . A  $C^\omega$ -manifold is termed to be a real analytic manifold. In case of  $k = 2m$ , then  $\mathbb{R}^{2m}$  is identified with  $\mathbb{C}^m$ , and the transition maps are required to be holomorphic.

**Definition 1.21.**  $M$  is a  $C^\infty$ -manifold or a *smooth manifold* if  $M$  is a topological manifold and there exists a family of charts  $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$  such that

1.  $\{U_\alpha\}_{\alpha \in \Lambda}$  is an open cover of  $M$ , and
2.  $(U_\alpha, \varphi_\alpha)$  and  $(U_\beta, \varphi_\beta)$  are  $C^\infty$ -compatible for all  $\alpha, \beta \in \Lambda$ .

Similarly,  $C^k$ -manifolds,  $C^\omega$ -manifolds, and complex manifolds may be defined. The family of charts  $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$  is called an *atlas* of  $M$ .

The most basic non-trivial example is perhaps the circle  $S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$ , which is a  $C^\infty$ -manifold of dimension 1. We can think of  $\mathbb{R}^2$  as  $\mathbb{C}$ . Let  $U = S^1 \setminus \{1\} = \{e^{it} \mid 0 < t < 2\pi\}$  and  $V = S^1 \setminus \{-1\} = \{e^{it} \mid -\pi < t < \pi\}$ . Then  $(U, \varphi)$  and  $(V, \psi)$  are charts, where  $\varphi : U \rightarrow (0, 2\pi) \subseteq \mathbb{R}$  is defined as  $e^{it} \mapsto t$  and  $\psi : V \rightarrow (-\pi, \pi) \subseteq \mathbb{R}$  is defined as  $e^{it} \mapsto t$ . Here,  $U \cap V = \{e^{it} \mid 0 < t < \pi\} \cup \{e^{-t} \mid \pi < t < 2\pi, -\pi < t < 0\}$ . The transition maps are  $\psi \circ \varphi^{-1} : (0, \pi) \rightarrow (0, \pi)$  defined as  $t \mapsto t$  and  $\psi \circ \varphi^{-1} : (\pi, 2\pi) \rightarrow (-\pi, 0)$  defined as  $t \mapsto t - 2\pi$ . Both these maps are smooth, and hence the charts are compatible. Thus,  $S^1$  is a smooth manifold of dimension 1.

*August 3rd.*

**Proposition 1.22.** Let  $\mathcal{U} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$  be an atlas on  $M$ . Suppose  $(V, \psi)$  and  $(W, \sigma)$  are two charts on  $M$  which are  $C^\infty$ -compatible with all charts in  $\mathcal{U}$ . Then  $(V, \psi)$  and  $(W, \sigma)$  are  $C^\infty$ -compatible.

*Proof.* Let  $p \in V \cap W$ . Then there exists  $\alpha \in \Lambda$  such that  $p \in U_\alpha$ . Since  $(V, \psi)$  and  $(U_\alpha, \varphi_\alpha)$  are compatible, we have that  $\psi \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap V) \rightarrow \psi(U_\alpha \cap V)$  is smooth. Similarly, since  $(W, \sigma)$  and  $(U_\alpha, \varphi_\alpha)$  are compatible, we have that  $\sigma \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap W) \rightarrow \sigma(U_\alpha \cap W)$  is smooth. Now, we have

$$(\sigma \circ \psi^{-1})|_{\psi(U_\alpha \cap V \cap W)} = (\sigma \circ \varphi_\alpha^{-1})|_{\varphi_\alpha(U_\alpha \cap V \cap W)} \circ (\varphi_\alpha \circ \psi^{-1})|_{\psi(U_\alpha \cap V \cap W)}. \quad (1.60)$$

The right-hand side is a composition of smooth maps, and hence is smooth. Thus,  $\sigma \circ \psi^{-1}$  is smooth on an open neighbourhood of  $p$ . Since  $p$  was arbitrary, we have that  $\sigma \circ \psi^{-1}$  is smooth on  $V \cap W$ . Similarly, we can show that  $\psi \circ \sigma^{-1}$  is smooth on  $V \cap W$ . Thus,  $(V, \psi)$  and  $(W, \sigma)$  are compatible. ■

This leads to the following definition.

**Definition 1.23.** A *maximal atlas*  $\mathcal{M}$  on a smooth manifold  $M$  is an atlas which is maximal with respect to inclusion; that is, if  $\mathcal{M} \subseteq \mathcal{U}$  for some atlas  $\mathcal{U}$ , then  $\mathcal{M} = \mathcal{U}$ . Equivalently,  $\mathcal{M}$  is a maximal atlas if it contains all charts which are compatible with all charts in  $\mathcal{M}$ .

Any atlas  $\mathcal{U}$  on  $M$  is contained in a unique maximal atlas  $\mathcal{M}$ . For a proof of this, let  $\mathcal{M} = \{(U, \varphi) \mid \text{compatible with all charts in } \mathcal{U}\}$ . Then  $\mathcal{M}$  is an atlas. If  $\mathcal{M} \subseteq \mathcal{M}'$ , then any chart in  $\mathcal{M}'$  must be compatible with all charts in  $\mathcal{M}$ ; it must be compatible with all charts in  $\mathcal{U}$ . By definition of  $\mathcal{M}$ , such a chart must be in  $\mathcal{M}$ , so  $\mathcal{M} = \mathcal{M}'$ . For uniqueness, if  $\mathcal{U} \subseteq \mathcal{M}, \mathcal{M}'$ , then  $\mathcal{M}, \mathcal{M}' \subseteq \mathcal{M} \cup \mathcal{M}'$  tells us  $\mathcal{M} = \mathcal{M}'$  by maximality. The unique maximal atlas  $\mathcal{M} \supseteq \mathcal{U}$  is also called a differentiable structure on  $M$ .

**Example 1.24.** 1. Let  $U \subseteq \mathbb{R}^n$ , and  $F : U \rightarrow \mathbb{R}^m$  be a  $C^\infty$ -map. The graph of  $F$  is defined as  $\Gamma(F) = \{(x, F(x)) \mid x \in U\} \subseteq \mathbb{R}^{n+m}$ . An atlas on  $\Gamma(F)$  is given by  $\{(U, \varphi)\}$ , where  $\varphi : U \rightarrow \Gamma(F)$  is defined as  $x \mapsto (x, F(x))$ . The transition map is  $\varphi^{-1} : \Gamma(F) \rightarrow U$  defined as  $(x, F(x)) \mapsto x$ . This is smooth, and hence  $\Gamma(F)$  is a smooth manifold of dimension  $n$ .

2. Let  $M$  and  $N$  be smooth manifolds of dimension  $m$  and  $n$ , respectively. Then  $M \times N$  is a smooth manifold of dimension  $m + n$ ; if  $\mathcal{U}$  and  $\mathcal{V}$  are atlases on  $M$  and  $N$ , respectively, then  $\mathcal{U} \times \mathcal{V} = \{(U \times V, \varphi \times \psi) \mid (U, \varphi) \in \mathcal{U}, (V, \psi) \in \mathcal{V}\}$  is an atlas on  $M \times N$ . The transition maps are  $(\varphi_2 \times \psi_2) \circ (\varphi_1 \times \psi_1)^{-1} = (\varphi_2 \circ \varphi_1^{-1}) \times (\psi_2 \circ \psi_1^{-1})$ , which is smooth since both  $\varphi_2 \circ \varphi_1^{-1}$  and  $\psi_2 \circ \psi_1^{-1}$  are smooth.

### 1.4.1 Smooth Functions and Smooth Maps

We will often omit ‘smooth’ but still mean smooth in the context of manifolds.

**Definition 1.25.** Let  $M$  be a manifold of dimension  $n$ . A function  $f : M \rightarrow \mathbb{R}$  is smooth at  $p \in M$  if there exists a chart  $(U, \varphi)$  of  $M$  such that  $f \circ \varphi^{-1} : \varphi(U) \rightarrow \mathbb{R}$  is smooth at  $\varphi(p)$ . A function  $f : M \rightarrow \mathbb{R}$  is smooth if it is smooth at all  $p \in M$ .

The definition of the smooth function  $f : M \rightarrow \mathbb{R}$  is independent of the choice of chart  $(U, \varphi)$ , since if  $(V, \psi)$  is another chart containing  $p$ , then  $\psi \circ \varphi^{-1} : \varphi(U \cap V) \rightarrow \psi(U \cap V)$  is smooth, and hence  $f \circ \psi^{-1} = (f \circ \varphi^{-1}) \circ (\varphi \circ \psi^{-1})$  is smooth at  $\psi(p)$ .

**Proposition 1.26.** Let  $M$  be a manifold of dimension  $n$ , and  $f : M \rightarrow \mathbb{R}$  be a function. Then the following are equivalent:

1.  $f$  is a smooth function on  $M$ .
2.  $M$  has an atlas  $\mathcal{U} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$  such that  $f \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha) \rightarrow \mathbb{R}$  is smooth for all  $\alpha \in \Lambda$ .
3. For all charts  $(U, \varphi)$  on  $M$ , the map  $f \circ \varphi^{-1} : \varphi(U) \rightarrow \mathbb{R}$  is smooth.

*Proof.* Note that 2. implies 1. by definition, and 3. implies 2. trivially. We show that 1. implies 3. Let  $(U, \varphi)$  be a chart on  $M$ , and  $p \in U$ ; it is enough to prove that  $f \circ \varphi^{-1}$  is smooth at  $\varphi(p)$ . Since  $f$  is smooth, there exists a chart  $(V, \psi)$  of  $M$  such that  $p \in V$  and  $f \circ \psi^{-1} : \psi(V) \rightarrow \mathbb{R}$  is smooth at  $\psi(p)$ . Now, we have

$$f \circ \varphi^{-1} = (f \circ \psi^{-1}) \circ (\psi \circ \varphi^{-1}). \quad (1.61)$$

The right-hand side is a composition of smooth maps, and hence is smooth at  $\varphi(p)$ . Thus,  $f \circ \varphi^{-1}$  is smooth at  $\varphi(p)$ . ■

A function  $f : M \rightarrow \mathbb{R}$  is smooth on a chart  $(U, \varphi)$  if and only if the pullback  $\varphi^* f := f \circ \varphi^{-1} : \varphi(U) \rightarrow \mathbb{R}$  is smooth. Also note that the definition of a smooth function  $f$  does not assume that  $f$  is continuous; however, one may show that the definition itself implies that  $f$  is continuous. This is because if  $f$  is smooth at  $p \in M$ , then there exists a chart  $(U, \varphi)$  of  $M$  such that  $f \circ \varphi^{-1} : \varphi(U) \rightarrow \mathbb{R}$  is smooth at  $\varphi(p)$ , and hence continuous at  $\varphi(p)$ . Since  $\varphi$  is a homeomorphism, we have that  $f$  is continuous at  $p$ . Since  $p$  was arbitrary, we have that  $f$  is continuous on  $M$ .

**Definition 1.27.** Let  $M$  and  $N$  be smooth manifolds of dimension  $m$  and  $n$ , respectively. A continuous map  $F : N \rightarrow M$  is smooth at  $p \in N$  if there exist charts  $(U, \varphi)$  on  $N$ , with  $p \in U$ , and  $(V, \psi)$  on  $M$ , with  $F(p) \in V$ , such that  $\psi \circ F \circ \varphi^{-1} : \varphi(U) \rightarrow \psi(V)$  is smooth at  $\varphi(p)$ . A continuous map  $F : N \rightarrow M$  is smooth if it is smooth at all  $p \in N$ .

The definition of a map  $f$  being smooth at  $p$  is independent of the choice of charts  $(U, \varphi)$  and  $(V, \psi)$ , since if  $(U', \varphi')$  and  $(V', \psi')$  are other charts containing  $p$  and  $F(p)$ , respectively, then  $\psi' \circ \psi^{-1} : \psi(V \cap V') \rightarrow \psi'(V \cap V')$  and  $\varphi' \circ \varphi^{-1} : \varphi(U \cap U') \rightarrow \varphi'(U \cap U')$  are smooth, and hence

$$\psi' \circ F \circ \varphi'^{-1} = (\psi' \circ \psi^{-1}) \circ (\psi \circ F \circ \varphi^{-1}) \circ (\varphi \circ \varphi'^{-1}) \quad (1.62)$$

**Proposition 1.28.** Let  $F : N \rightarrow M$  be a continuous map between smooth manifolds. Then the following are equivalent:

1.  $F$  is smooth.
2. There exist atlases  $\mathcal{U}$  of  $N$  and  $\mathcal{V}$  of  $M$  such that for all  $(U, \varphi) \in \mathcal{U}$  and  $(V, \psi) \in \mathcal{V}$ , the map  $\psi \circ F \circ \varphi^{-1} : \varphi(U \cap F^{-1}(V)) \rightarrow \psi(V)$  is smooth.
3. For all charts  $(U, \varphi)$  on  $N$  and  $(V, \psi)$  on  $M$ , the map  $\psi \circ F \circ \varphi^{-1} : \varphi(U \cap F^{-1}(V)) \rightarrow \psi(V)$  is smooth.

*Proof.* Again, note that 2. implies 1. by definition, and 3. implies 2. trivially. A similar argument as in the previous proposition shows that 1. implies 3. ■

**Proposition 1.29.** *Let  $F : N \rightarrow M$  and  $G : M \rightarrow P$  be smooth maps between smooth manifolds. Then the composition  $G \circ F : N \rightarrow P$  is smooth.*

*Proof.* Let  $(U, \varphi)$ ,  $(V, \psi)$ , and  $(W, \sigma)$  be charts on  $N$ ,  $M$ , and  $P$ , respectively, such that  $p \in U$ ,  $F(p) \in V$ , and  $G(F(p)) \in W$ . Then we have

$$\sigma \circ (G \circ F) \circ \varphi^{-1} = (\sigma \circ G \circ \psi^{-1}) \circ (\psi \circ F \circ \varphi^{-1}). \quad (1.63)$$

The right-hand side is a composition of smooth maps, and hence is smooth at  $\varphi(p)$ . Thus,  $G \circ F$  is smooth at  $p$ . ■

Finally, a smooth map  $F : N \rightarrow M$  is termed a *diffeomorphism* if  $F$  is a bijection, and  $F$  and  $F^{-1}$  are smooth. Some properties of diffeomorphisms may be inferred.

**Proposition 1.30.** *The following hold true.*

1. *Let  $(U, \varphi)$  be a chart on a smooth manifold  $M$  of dimension  $n$ . Then  $\varphi : U \rightarrow \varphi(U)$  is a diffeomorphism.*
2. *Suppose  $U \subseteq M$  is open, and that  $F : U \rightarrow F(U)$  is a diffeomorphism. Then  $(U, F)$  is a chart in the maximal atlas of  $M$ .*

*Proof.* 1. Note that  $U$  is a manifold since it is open in  $M$ , and  $\varphi(U)$  is a manifold since it is open in  $\mathbb{R}^n$ . The map  $\varphi : U \rightarrow \varphi(U)$  is a homeomorphism by definition of a chart, so it is bijective. Note that  $\{(U, \varphi)\}$  is an atlas on  $U$ , and  $\{(\varphi(U), \text{id}_{\varphi(U)})\}$  is an atlas on  $\varphi(U)$ . Then  $\varphi$  is smooth since  $\text{id}_{\varphi(U)} \circ \varphi \circ \varphi^{-1} = \text{id}_{\varphi(U)}$  is smooth, and  $\varphi^{-1}$  is smooth since  $\varphi \circ \varphi^{-1} \circ \text{id}_{\varphi(U)} = \text{id}_{\varphi(U)}$  is smooth. Thus,  $\varphi$  is a diffeomorphism.

2. Since  $U$  is open in  $M$ , it is a smooth manifold. Because  $F : U \rightarrow F(U)$  is a diffeomorphism, it is in particular a homeomorphism. Hence  $F$  is bijective, and its image  $F(U)$  is an open subset of  $\mathbb{R}^n$  (as required for a chart). Let  $(V, \psi)$  be any chart on  $M$  with  $U \cap V \neq \emptyset$ . Then the transition maps

$$\psi \circ F^{-1} : F(U \cap V) \rightarrow \psi(V) \quad \text{and} \quad F \circ \psi^{-1} : \psi(U \cap V) \rightarrow F(U \cap V) \quad (1.64)$$

are smooth, since they are compositions of smooth maps. Therefore  $(U, F)$  is compatible with every chart in the maximal atlas of  $M$ , and so  $(U, F)$  is a chart in that maximal atlas. ■

**Proposition 1.31.** *Let  $N$  be a smooth manifold of dimension  $n$ , and  $F : N \rightarrow \mathbb{R}^m$  be a continuous map. Then the following are equivalent.*

1.  *$F$  is smooth.*
2.  *$N$  has an atlas  $\mathcal{U}$  such that for all  $(U, \varphi) \in \mathcal{U}$ , the map  $F \circ \varphi^{-1} : \varphi(U) \rightarrow \mathbb{R}^m$  is smooth.*
3. *For all charts  $(U, \varphi)$  on  $N$ , the map  $F \circ \varphi^{-1} : \varphi(U) \rightarrow \mathbb{R}^m$  is smooth.*

*Proof.* 3. implies 2. is obvious. For 2. implies 1., note that there exist atlases  $\mathcal{U}$  and  $\mathcal{V}$  of  $N$  and  $M$  respectively such that  $(U, \varphi) \in \mathcal{U}$  and  $(V, \psi) \in \mathcal{V}$  satisfy the smoothness condition. Since  $\mathbb{R}^m$  has a global chart  $(\mathbb{R}^m, \text{id}_{\mathbb{R}^m})$ , we can take  $\mathcal{V} = \{(\mathbb{R}^m, \text{id}_{\mathbb{R}^m})\}$ . Then for any  $(U, \varphi) \in \mathcal{U}$ , we have  $\text{id}_{\mathbb{R}^m} \circ F \circ \varphi^{-1} = F \circ \varphi^{-1}$  is smooth, which implies that  $F$  is smooth. For 1. implies 3.,  $F$  is smooth and  $\varphi$  is a diffeomorphism; thus,  $\varphi^{-1}$  is smooth, and  $F \circ \varphi^{-1}$  is smooth. ■

**Proposition 1.32.** *Let  $F : N \rightarrow M$  be a continuous map between smooth manifolds. Then the following are equivalent.*

1.  *$F$  is smooth.*

2.  $M$  has an atlas  $\mathcal{V}$  such that for all  $(V, \psi) \in \mathcal{V}$ , the map  $\psi \circ F : F^{-1}(V) \rightarrow \psi(V)$  is smooth.
3. For any chart  $(V, \psi)$  of  $M$ , the map  $\psi \circ F : F^{-1}(V) \rightarrow \psi(V)$  is smooth.

The proof follows a similar structure to the one above.

**Proposition 1.33.** *Let  $F : N \rightarrow \mathbb{R}^m$  be a continuous map. Then the following are equivalent.*

1.  $F$  is smooth.
2.  $F^1, \dots, F^m : N \rightarrow \mathbb{R}$  are smooth, where  $F = (F^1, \dots, F^m)$ .

## Examples

August 5th.

We look at some examples of smooth manifolds and smooth maps.

**Example 1.34.** For  $S^n \subseteq \mathbb{R}^{n+1}$ , the  $n$ -sphere, define  $U_i^+ = \{(x_1, \dots, x_{n+1}) \in S^n \mid x_i > 0\}$  and  $U_i^- = \{(x_1, \dots, x_{n+1}) \in S^n \mid x_i < 0\}$  for  $i = 1, \dots, n+1$ . Then  $\{(U_i^+, \varphi_i^+), (U_i^-, \varphi_i^-)\}_{i=1}^{n+1}$  is an atlas on  $S^n$ , where  $\varphi_i^+ : U_i^+ \rightarrow \mathbb{R}^n$  is defined as  $(x_1, \dots, x_{n+1}) \mapsto (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_{n+1})$ , and  $\varphi_i^- : U_i^- \rightarrow \mathbb{R}^n$  is defined similarly. The transition maps are smooth, and hence  $S^n$  is a smooth manifold of dimension  $n$ .

**Example 1.35.** Recall  $\mathbb{R}P^n$ , the *real projective space* of dimension  $n$ , defined as the quotient  $\mathbb{R}^{n+1} \setminus \{0\} / \sim$ , where  $(x_1, \dots, x_{n+1}) \sim (y_1, \dots, y_{n+1})$  if there exists  $\lambda \in \mathbb{R} \setminus \{0\}$  such that  $(x_1, \dots, x_n) = (\lambda y_1, \dots, \lambda y_n)$ . Let  $(a_1 : a_2 : \dots : a_{n+1})$  denote the equivalence class of  $(a_1, \dots, a_{n+1}) \in \mathbb{R}^{n+1} \setminus \{0\}$ . This is a smooth manifold of dimension  $n$ ; an atlas is given by  $\{(U_i, \varphi_i)\}_{i=1}^{n+1}$ , where  $U_i = \{(x_1 : \dots : x_{n+1}) \in \mathbb{R}P^n \mid x_i \neq 0\}$  and  $\varphi_i : U_i \rightarrow \mathbb{R}^n$  is defined as

$$\varphi_i(x_1 : \dots : x_{n+1}) = \left( \frac{x_1}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_{n+1}}{x_i} \right). \quad (1.65)$$

Verify that the transition maps  $\varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \rightarrow \varphi_j(U_i \cap U_j)$  are smooth for all  $i, j = 1, \dots, n+1$ . We can also provide a smooth map  $F : S^n \rightarrow \mathbb{R}P^n$  defined as  $(x_1, \dots, x_{n+1}) \mapsto (x_1 : \dots : x_{n+1})$ . This is a smooth map, and it is a 2-to-1 covering map.

**Example 1.36.** Similar to the above, we can define  $\mathbb{P}_k^n$  to be the projective space of dimension  $n$  over the field  $k$ ; that is,  $\mathbb{P}_k^n = k^{n+1} \setminus \{0\} / \sim$ , where  $(x_1, \dots, x_{n+1}) \sim (y_1, \dots, y_{n+1})$  if there exists  $\lambda \in k \setminus \{0\}$  such that  $(x_1, \dots, x_{n+1}) = (\lambda y_1, \dots, \lambda y_{n+1})$ . In particular, if  $k = \mathbb{C}$ , then  $\mathbb{P}_{\mathbb{C}}^n$  consists of complex 1-dimensional subspaces of  $\mathbb{C}^{n+1}$ . This is a smooth manifold of dimension  $2n$ .  $\mathbb{R}P^n$  may be, thus, interpreted as the space of real 1-dimensional subspaces of  $\mathbb{R}^{n+1}$ . Building upon this, the *Grassmannian space*  $\text{Gr}(n, k)$  is defined as the space of  $k$ -dimensional subspaces of  $\mathbb{R}^n$ . This is a smooth manifold of dimension  $k(n-k)$ . The Grassmannian space  $\text{Gr}(n+1, 1)$  is precisely  $\mathbb{R}P^n$ , and  $\text{Gr}(n, 0)$  is a single point.  $\text{Gr}(n, n-1)$  is the space of hyperplanes in  $\mathbb{R}^n$ . Again, the Grassmannian construction may be generalized to any field.

## 1.4.2 Quotient Manifolds

August 10th.

Recall that for an equivalence relation  $\sim$  on a set  $S$ , the quotient set  $S/\sim$  is defined as the set of equivalence classes  $[s] = \{t \in S \mid t \sim s\} \in S/\sim$ . If  $S$  happens to be a topological space, then the projection map  $\pi : S \rightarrow S/\sim$  defined as  $s \mapsto [s]$  is continuous if we endow  $S/\sim$  with the quotient topology; that is, a subset  $U \subseteq S/\sim$  is open if and only if  $\pi^{-1}(U) \subseteq S$  is open. This is the finest topology on  $S/\sim$  which makes  $\pi$  continuous. Also recall the universal property; that is, if  $f : S \rightarrow X$  is a set map between topological spaces, and  $\pi : S \rightarrow S/\sim$  is the projection map, then there exists a unique map  $\tilde{f} : S/\sim \rightarrow X$

such that  $f = \tilde{f} \circ \pi$  if and only if  $f$  is constant on equivalence classes. In this case,  $\tilde{f}$  is continuous if and only if  $f$  is continuous.

The equivalence relation  $\sim$  on  $S$  is called open if the projection map  $\pi : S \rightarrow S/\sim$  is open; that is, if  $U \subseteq S$  is open, then  $\pi(U) \subseteq S/\sim$  is open. Usually,  $S/\sim$  is not  $T_2$  is general. The following is a sufficient condition for  $S/\sim$  to be  $T_2$ , or Hausdorff.

**Proposition 1.37.** *Let  $\sim$  be an open equivalence relation on a topological space  $S$ . Then  $S/\sim$  is  $T_2$  if and only if  $\{(x, y) \in S \times S \mid x \sim y\}$  is closed in  $S \times S$ .*

*Proof.* Suppose the set  $R = \{(x, y) \in S \times S \mid x \sim y\}$  is closed in  $S \times S$ . That is, for all  $(x, y) \in S \times S$  such that  $x \not\sim y$ , there exist open neighbourhoods  $U$  of  $x$  and  $V$  of  $y$  such that  $U \times V \subseteq S \times S \setminus R$ . Since  $\sim$  is open, we have that  $\pi(U)$  and  $\pi(V)$  are open neighbourhoods of  $[x]$  and  $[y]$ , respectively. We claim that  $\pi(U) \cap \pi(V) = \emptyset$ . Suppose not; then there exists  $[z] \in \pi(U) \cap \pi(V)$ . Then there exist  $u \in U$  and  $v \in V$  such that  $[z] = [u] = [v]$ , which implies that  $u \sim v$ . But this contradicts the fact that  $(u, v) \in U \times V \subseteq S \times S \setminus R$ . Thus,  $\pi(U) \cap \pi(V) = \emptyset$ , and hence  $S/\sim$  is Hausdorff.

On the other hand, suppose  $S/\sim$  is Hausdorff. Let  $(x, y) \in S \times S$  such that  $x \not\sim y$ . Then there exist open neighbourhoods  $U$  of  $[x]$  and  $V$  of  $[y]$  in  $S/\sim$  such that  $U \cap V = \emptyset$ . Since  $\pi : S \rightarrow S/\sim$  is continuous, we have that  $\pi^{-1}(U)$  and  $\pi^{-1}(V)$  are open neighbourhoods of  $x$  and  $y$ , respectively. We claim that  $\pi^{-1}(U) \times \pi^{-1}(V) \subseteq S \times S \setminus R$ . Suppose not; then there exists  $(u, v) \in \pi^{-1}(U) \times \pi^{-1}(V)$  such that  $u \sim v$ . Then we have  $[u] = [v]$ , which implies that  $[u] \in U \cap V$ , contradicting the fact that  $U \cap V = \emptyset$ . Thus,  $\pi^{-1}(U) \times \pi^{-1}(V) \subseteq S \times S \setminus R$ , and hence  $R$  is closed in  $S \times S$ . ■

**Example 1.38.** Let  $\mathbb{R}^{n+1} \setminus \{(0, \dots, 0)\}$  be the punctured space, and define an equivalence relation  $\sim$  on it as  $(x_1, \dots, x_{n+1}) \sim (y_1, \dots, y_{n+1})$  if there exists  $\lambda \in \mathbb{R} \setminus \{0\}$  such that  $(x_1, \dots, x_{n+1}) = (\lambda y_1, \dots, \lambda y_{n+1})$ . Then  $\mathbb{R}^{n+1} \setminus \{(0, \dots, 0)\} / \sim$  is precisely  $\mathbb{R}P^n$ . The projection map  $\pi : \mathbb{R}^{n+1} \setminus \{(0, \dots, 0)\} \rightarrow \mathbb{R}P^n$  is open; let  $U \subseteq \mathbb{R}^n \setminus \{(0, \dots, 0)\}$  be open. Then  $\pi^{-1}(\pi(U)) = \bigcup_{\lambda \in \mathbb{R} \setminus \{0\}} \lambda U$  is open, and hence  $\pi(U)$  is open in  $\mathbb{R}P^n$ . Also, the set  $R = \{(x, y) \in (\mathbb{R}^{n+1} \setminus \{0\}) \times (\mathbb{R}^{n+1} \setminus \{0\}) \mid x \sim y\}$  is closed in  $(\mathbb{R}^{n+1} \setminus \{0\}) \times (\mathbb{R}^{n+1} \setminus \{0\})$ , since if  $y_i = \lambda x_i$  for some  $\lambda \in \mathbb{R} \setminus \{0\}$ , then  $x_i y_j - x_j y_i = 0$  for all  $i, j = 1, \dots, n+1$ . Thus,  $R$  is the zero set of the continuous functions  $f_{ij} : (\mathbb{R}^{n+1} \setminus \{0\}) \times (\mathbb{R}^{n+1} \setminus \{0\}) \rightarrow \mathbb{R}$  defined as  $(x, y) \mapsto x_i y_j - x_j y_i$  for all  $i, j = 1, \dots, n+1$ . Hence,  $R$  is closed in  $(\mathbb{R}^{n+1} \setminus \{0\}) \times (\mathbb{R}^{n+1} \setminus \{0\})$ . By the previous proposition, we have that  $\mathbb{R}P^n$  is Hausdorff.

**Proposition 1.39.** *Let  $\sim$  be an open equivalence relation on a topological space  $S$ . Then  $S$  is second countable if and only if  $S/\sim$  is second countable.*

*Proof.* Suppose  $S$  is second countable, and let  $\mathcal{B}$  be a countable basis for  $S$ . Then  $\mathcal{B}' = \{\pi(B) \mid B \in \mathcal{B}\}$  is a countable collection of open sets in  $S/\sim$ . We claim that  $\mathcal{B}'$  is a basis for  $S/\sim$ . Let  $U \subseteq S/\sim$  be open, and let  $[x] \in U$ . Then  $\pi^{-1}(U) \subseteq S$  is open, and hence there exists  $B \in \mathcal{B}$  such that  $x \in B \subseteq \pi^{-1}(U)$ . Then we have  $[x] \in \pi(B) \subseteq U$ , and hence  $\mathcal{B}'$  is a basis for  $S/\sim$ . Thus,  $S/\sim$  is second countable.

On the other hand, suppose  $S/\sim$  is second countable, and let  $\mathcal{B}'$  be a countable basis for  $S/\sim$ . Then  $\mathcal{B} = \{\pi^{-1}(B') \mid B' \in \mathcal{B}'\}$  is a countable collection of open sets in  $S$ . We claim that  $\mathcal{B}$  is a basis for  $S$ . Let  $U \subseteq S$  be open, and let  $x \in U$ . Then  $\pi(U) \subseteq S/\sim$  is open, and hence there exists  $B' \in \mathcal{B}'$  such that  $[x] \in B' \subseteq \pi(U)$ . Then we have  $x \in \pi^{-1}(B') \subseteq U$ , and hence  $\mathcal{B}$  is a basis for  $S$ . Thus,  $S$  is second countable. ■

**Example 1.40.** Consider the disc  $D^n = \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_1^2 + \dots + x_n^2 \leq 1\}$ , and the hemisphere  $H^n = \{(x_1, \dots, x_{n+1}) \in S^n \mid x_{n+1} \geq 0\}$ . On  $H^n$ , define an equivalence relation as  $(x_1, \dots, x_n, x_{n+1}) \sim (y_1, \dots, y_n, y_{n+1})$  if either

1.  $x_{n+1} = y_{n+1} = 0$  and  $(y_1, \dots, y_n, 0) = (-x_1, \dots, -x_n, 0)$ , or
2.  $(x_1, \dots, x_n, x_{n+1}) = (y_1, \dots, y_n, y_{n+1})$ .

On  $D^n$ , define an equivalence relation simply as  $x \sim y$  if  $x = y$  or  $x = -y$ . Then, one can show that  $\mathbb{R}P^n \cong S^n / \sim \cong D^n / \sim \cong H^n / \sim$  as smooth manifolds, where  $\sim$  on  $S^n$  is the antipodal equivalence relation.

## 1.5 Differential Structure

Let  $M$  be a smooth manifold of dimension  $n$ . For a point  $p \in M$ , let  $(U, \varphi)$  be a chart of  $M$  such that  $p \in U$ . The functions  $r^1, \dots, r^n : \mathbb{R}^n \rightarrow \mathbb{R}$  defined as  $r^i(x_1, \dots, x_n) = x_i$  are smooth, and hence, we can interpret  $(U, \varphi)$  as  $(U, x^1, \dots, x^n)$ , a chart with smooth coordinate functions  $x^i = r^i \circ \varphi : U \rightarrow \mathbb{R}$  for  $i = 1, \dots, n$ . Then, given a map  $f \in C^\infty(U)$ , we define the partial derivative of  $f$  with respect to the coordinate function  $x^i$  at  $p$  as

$$\frac{\partial f}{\partial x^i}(p) := \frac{\partial(f \circ \varphi^{-1})}{\partial r^i}(\varphi(p)). \quad (1.66)$$

In fact, given any  $p$  in this particular  $U$ , we have

$$(\varphi^{-1})^* \left( \frac{\partial f}{\partial x^i} \right) = \frac{\partial f}{\partial x^i} \circ \varphi^{-1} = \frac{\partial(f \circ \varphi^{-1})}{\partial r^i} \in C^\infty(\varphi(U)). \quad (1.67)$$

This implies that  $\frac{\partial f}{\partial x^i} \in C^\infty(U)$ , and hence the partial derivative of  $f$  with respect to  $x^i$  is smooth on  $U$ . Moreover, one can show that  $\frac{\partial x^i}{\partial x^j}(p) = \frac{\partial r^i}{\partial r^j}(\varphi(p)) = \delta_j^i$ .

Let  $F : N \rightarrow M$  be a smooth map between smooth manifolds of dimension  $n$  and  $m$ , respectively. Let  $(U, \varphi)$  be a chart on  $N$  with  $p \in U$ , and  $(V, \psi)$  be a chart on  $M$  with  $F(p) \in V$ . Then we can interpret  $(U, \varphi)$  as  $(U, x^1, \dots, x^n)$ , and  $(V, \psi)$  as  $(V, y^1, \dots, y^m)$ , where  $x^i = r^i \circ \varphi : U \rightarrow \mathbb{R}$  and  $y^j = r^j \circ \psi : V \rightarrow \mathbb{R}$  are smooth coordinate functions. Then, we have  $F^i = y^i \circ F = r^i \circ \psi \circ F \in C^\infty(U)$  for  $i = 1, \dots, m$ . The *Jacobian matrix* of  $F$  at  $p$ , with respect to the charts  $(U, \varphi)$  and  $(V, \psi)$ , is defined as the  $m \times n$  matrix

$$J_F(p) = \left( \frac{\partial F^i}{\partial x^j}(p) \right)_{1 \leq i \leq m, 1 \leq j \leq n}. \quad (1.68)$$

Each entry here can be written as

$$\frac{\partial F^i}{\partial x^j}(p) = \frac{\partial(\psi \circ F \circ \varphi^{-1})^i}{\partial r^j}(\varphi(p)) = \frac{\partial(r^i \psi F \varphi^{-1})}{\partial r^j}(\varphi(p)) = \frac{\partial(y^i F \varphi^{-1})}{\partial r^j}(\varphi(p)) = \frac{\partial(F^i \circ \varphi^{-1})}{\partial r^j}(\varphi(p)). \quad (1.69)$$

Recall the inverse function theorem, which states that if  $F : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$  is smooth, and the Jacobian matrix  $J_F(p)$  is invertible at  $p \in U$ , then there exists an open neighbourhood  $V$  of  $F(p)$  such that  $F : U \rightarrow V$  is a diffeomorphism. This motivates the following definition. Thus, if  $m = n$  and  $J_F(p)$  is invertible, then  $F$  is a local diffeomorphism at  $p$ ;  $\psi \circ F \circ \varphi^{-1}$  is a local diffeomorphism at  $\varphi(p)$  if and only if  $F$  is a local diffeomorphism at  $p$ .

Let  $M$  be a smooth manifold of dimension  $n$ , and let  $p \in M$ . Suppose  $F^1, F^2, \dots, F^n$  are smooth functions on some open neighbourhood  $U \subseteq M$  of  $p$ . Then consider  $F = (F^1, \dots, F^n) : U \rightarrow \mathbb{R}^n$ , which is a smooth map. Then  $\left( \frac{\partial F^i}{\partial x^j}(p) \right)_{n \times n}$  is invertible if and only if  $F$  is a local diffeomorphism at  $p$  if and only if  $F : W \rightarrow F(W) \subseteq \mathbb{R}^n$  is a diffeomorphism for some open neighbourhood  $W \subseteq U$  of  $p$ . Then, on some open set around  $p$ , we can use  $F^1, \dots, F^n$  as local coordinates. This motivates the following definition.

**Definition 1.41.** Let  $M$  be a smooth manifold, with  $p \in M$ . Recall  $C_p^\infty(M) \equiv C_p^\infty$ , the local ring of germs of smooth functions at  $p$ , which is also an  $\mathbb{R}$ -algebra. Also define  $T_p M$  to be the  $\mathbb{R}$ -vector space of derivations at  $p$ ; that is,  $T_p M = \{D : C_p^\infty \rightarrow \mathbb{R} \mid D \text{ is linear and satisfies the Leibniz rule}\}$ . This  $T_p M$  is called the *tangent space* of  $M$  at  $p$ .

August 12th.

We leave it as an exercise to check that  $\frac{\partial}{\partial x^i}|_p$  is a derivation from  $C_p^\infty$  to  $\mathbb{R}$ , and hence  $\frac{\partial}{\partial x^i}|_p \in T_p M$  for all valid  $i$ . Given a smooth map  $F : N \rightarrow M$  between smooth manifolds, we can define the map  $F_{*,p} : T_p N \rightarrow T_{F(p)} M$  as  $D \mapsto F_{*,p}(D)$ , where  $F_{*,p}(D)(f) = D(f \circ F)$  for all  $f \in C_{F(p)}^\infty$ . This is the pushforward of  $f$  at  $p$ . Moreover, if  $G : M \rightarrow P$  is another smooth map, then  $(G \circ F)_{*,p} = G_{*,F(p)} \circ F_{*,p}$ . The map  $F_{*,p}$  is linear, and hence a linear transformation between the tangent spaces  $T_p N$  and  $T_{F(p)} M$ .

If  $F : N \rightarrow M$  were a diffeomorphism, then there exists a diffeomorphism  $G : M \rightarrow N$  such that  $G \circ F = \text{id}_N$  and  $F \circ G = \text{id}_M$ . Now look at the chain

$$T_p N \xrightarrow{F_{*,p}} T_{F(p)} M \xrightarrow{G_{*,F(p)}} T_{G(F(p))} N = T_p N. \quad (1.70)$$

Here, we have  $(G \circ F)_{*,p} = G_{*,F(p)} \circ F_{*,p} = (\text{id}_N)_{*,p} = \text{id}_{T_p N}$ , and similarly  $F_{*,p} \circ G_{*,F(p)} = \text{id}_{T_{F(p)} M}$ . Thus,  $F_{*,p}$  is a linear isomorphism between the tangent spaces  $T_p N$  and  $T_{F(p)} M$ .

Let  $p \in M$ , a smooth manifold, and let  $(U, \varphi)$  be a chart containing  $p$ . Since  $C_p^\infty(M) \equiv C_p^\infty(U)$ , the diffeomorphism  $\varphi : U \rightarrow \varphi(U)$  induces an isomorphism  $\varphi_{*,p} : T_p M \rightarrow T_{\varphi(p)} \mathbb{R}^n$ . We know that the tangent space on the right has the basis  $\{\frac{\partial}{\partial x^1}|_{\varphi(p)}, \dots, \frac{\partial}{\partial x^n}|_{\varphi(p)}\}$ . Applying this map to a basis element, we have

$$\varphi_{*,p} \left( \frac{\partial}{\partial x^i} \Big|_p \right) (f) = \frac{\partial}{\partial x^i} \Big|_p (f \circ \varphi) = \frac{\partial}{\partial r^i} \Big|_{\varphi(p)} (f \circ \varphi \circ \varphi^{-1}) = \frac{\partial}{\partial r^i} \Big|_{\varphi(p)} (f) \quad \text{for all } f \in C_p^\infty(M). \quad (1.71)$$

Hence,  $\varphi_{*,p} \left( \frac{\partial}{\partial x^i} \Big|_p \right) = \frac{\partial}{\partial r^i} \Big|_{\varphi(p)}$ , and so  $\{\frac{\partial}{\partial x^1}|_p, \dots, \frac{\partial}{\partial x^n}|_p\}$  is a basis for  $T_p M$ . Thus, we have shown that the tangent space  $T_p M$  has dimension  $n$ . This also shows that if  $F : N \rightarrow M$  is a smooth map between smooth manifolds of dimension  $n$  and  $m$  that also happens to be a diffeomorphism, then  $n = m$ . Moreover, given a map  $F : N \rightarrow M$  between smooth manifolds, and given a chart  $(U, \varphi)$  of  $p \in N$  and a chart  $(V, \psi)$  of  $F(p) \in M$ , we have

$$F_{*,p} : T_p N \rightarrow T_{F(p)} M, \quad F_{*,p} \left( \frac{\partial}{\partial x^i} \Big|_p \right) = \sum_{j=1}^m b_j^i \frac{\partial}{\partial y^j} \Big|_{F(p)}, \quad b_j^i = \frac{\partial F^j}{\partial x^i}(p) \quad \text{for } i = 1, \dots, n, j = 1, \dots, m. \quad (1.72)$$

That is, the matrix of  $F_{*,p}$  with respect to the bases  $\{\frac{\partial}{\partial x^1}|_p, \dots, \frac{\partial}{\partial x^n}|_p\}$  and  $\{\frac{\partial}{\partial y^1}|_{F(p)}, \dots, \frac{\partial}{\partial y^m}|_{F(p)}\}$  is precisely the  $m \times n$  real *Jacobian matrix*  $(dF)_p := J_F(p) = \left( \frac{\partial F^j}{\partial x^i}(p) \right)_{1 \leq i \leq n, 1 \leq j \leq m}$ .

*August 17th.*

Suppose  $0 \in (a, b) \subseteq \mathbb{R}$ , and let  $c : (a, b) \rightarrow M$  be a smooth map to a smooth manifold  $M$  of dimension  $n$  such that  $c(0) = p \in M$ .  $c$  is then called a *smooth curve* in  $M$ . Let  $(U, \varphi) = (U, x^1, \dots, x^n)$  be a chart on  $M$  with  $p \in U$ . Then  $c'(0) \in T_p M$  is defined as  $c'(0) = c_{*,0} \left( \frac{d}{dt} \Big|_0 \right)$ , where  $\frac{d}{dt} \Big|_0 \in T_0 \mathbb{R}$  is the standard basis element. Then  $c'(0)$  is a tangent vector at  $p$ , and we have

$$\mathbb{R} \cong T_0((a, b)) \xrightarrow{c_{*,0}} T_p M \xrightarrow{\varphi_{*,p}} T_{\varphi(p)} \mathbb{R}^n \cong \mathbb{R}^n, \quad c'(0) \mapsto \varphi_{*,p}(c'(0)) = (\varphi \circ c)_{*,0} \left( \frac{d}{dt} \Big|_0 \right) = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i} \Big|_p \quad (1.73)$$

since  $\{\frac{\partial}{\partial x^i}|_p\}_{i=1}^n$  is a basis for  $T_p M$ . Utilising  $c'(0) = c_{*,0} \left( \frac{d}{dt} \Big|_0 \right)$ , we have

$$a^i = \varphi_{*,p}(c'(0))(x^i) = c_{*,0} \left( \frac{d}{dt} \Big|_0 \right) (x^i \circ \varphi^{-1}) = \frac{d}{dt} \Big|_0 (x^i \circ \varphi^{-1} \circ c) = \frac{d}{dt} \Big|_0 (x^i \circ c) = \frac{dc^i}{dt}(0). \quad (1.74)$$

Thus, a curve  $c$  can be thought of as a tangent vector at  $p$ , and the tangent vector is given by the derivative of the curve at 0. In particular, let us interpret  $c : \mathbb{R} \rightarrow \mathbb{R}^n$  as a smooth curve in  $\mathbb{R}^n$ ; we have  $c(t) = (c_1(t), \dots, c_n(t))$ , and hence  $c'(0) = (\frac{dc_1}{dt}(0), \dots, \frac{dc_n}{dt}(0)) \in \mathbb{R}^n$ . This is consistent with the above definition of  $c'(0)$  as a tangent vector at  $p$ . From here, we wish to obtain a smooth map  $c : (a, b) \rightarrow M$  such that  $c(0) = p$  and  $c'(0) = X_p = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i} \Big|_p$  for some  $X_p \in T_p M$ . Let  $(U, \varphi) = (U, x^1, \dots, x^n)$  be a chart on  $M$  with  $p \in U$ . In fact, choose a chart  $\varphi$  such that  $\varphi(p) = 0 \in \mathbb{R}^n$  (translations are diffeomorphisms). Then we can define a smooth curve  $c : (-\epsilon, \epsilon) \rightarrow M$  for some  $\epsilon > 0$  such that  $c(0) = p$  and  $c'(0) = X_p$  as

$$c(t) = \varphi^{-1}(ta^1, \dots, ta^n) \quad \text{for } t \in (-\epsilon, \epsilon). \quad (1.75)$$

Then,  $c'(0) = c_{*,0} \left( \frac{d}{dt} \Big|_0 \right) = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i} \Big|_p = X_p$ , as desired.

Coming back to the smooth map  $F : N \rightarrow M$  between smooth manifolds, the rank of  $F$  at  $p \in N$  is defined as the rank of the linear map  $F_{*,p} : T_p N \rightarrow T_{F(p)} M$ , which is also the rank of the matrix  $(dF)_p = J_F(p) = \left( \frac{\partial F^j}{\partial x^i}(p) \right)_{1 \leq i \leq n, 1 \leq j \leq m}$ . The map  $F : N \rightarrow M$  is called an *immersion* at  $p \in N$  if  $F_{*,p} : T_p N \rightarrow T_{F(p)} M$  is injective (that is,  $\text{rank}(F_{*,p}) = \dim(T_p N) = n$ ), and  $F$  is called an immersion if it is an immersion at all points  $p \in N$ . Similarly,  $F$  is called a *submersion* at  $p \in N$  if  $F_{*,p} : T_p N \rightarrow T_{F(p)} M$  is surjective (that is,  $\text{rank}(F_{*,p}) = \dim(T_{F(p)} M) = m$ ), and  $F$  is called a submersion if it is a submersion at all points  $p \in N$ . If  $F : N \rightarrow M$  is a smooth map between smooth manifolds of dimension  $n$  and  $m$ , respectively, then  $\text{rank}(F_{*,p}) \leq \min\{n, m\}$  for all  $p \in N$ . Thus, if  $n < m$ , then  $F$  cannot be a submersion, and if  $n > m$ , then  $F$  cannot be an immersion.

## 1.6 Regular Maps and Regular Submanifolds

**Definition 1.42.** Let  $F : N \rightarrow M$  be a smooth map between smooth manifolds.  $p \in N$  is called a *critical point* of  $F$  if  $F_{*,p} : T_p N \rightarrow T_{F(p)} M$  is not surjective; that is,  $F$  is not a submersion at  $p$ .  $p$  is called a *regular point* of  $F$  if  $F_{*,p} : T_p N \rightarrow T_{F(p)} M$  is surjective; that is,  $F$  is a submersion at  $p$ . Similarly,  $q \in M$  is called a *critical value* of  $F$  if there exists a critical point  $p \in N$  such that  $F(p) = q$ .  $q \in M$  is called a *regular value* of  $F$  if it is not a critical value; that is, for all  $p \in N$  such that  $F(p) = q$ , we have that  $p$  is a regular point of  $F$ . That is, either  $F^{-1}(m) = \emptyset$ , or  $F_{*,p} : T_p N \rightarrow T_{F(p)} M$  is surjective for all  $p \in F^{-1}(m)$ .

From here, we can define something called a regular submanifold, which is useful in determining whether a certain set is a manifold or not. Let  $M$  be a manifold of dimension  $n$ , and  $S \subseteq M$  be a subset with the subspace topology. Suppose that for all  $p \in S$ , there exists a chart  $(U, \varphi) = (U, x^1, \dots, x^n)$  of  $M$  with  $p \in U$  and  $\varphi(p) = 0 \in \varphi(U) \subseteq \mathbb{R}^n$  such that

$$U \cap S = \{(x^1, x^2, \dots, x^n) \in U \mid x^{i_1} = x^{i_2} = \dots = x^{i_{n-k}} = 0 \text{ for some } 1 \leq i_1 < i_2 < \dots < i_{n-k} \leq n\} \quad (1.76)$$

for some fixed integer  $0 \leq k \leq n$  (via switching of coordinates, we may assume that the last  $n - k$  coordinates are zero). Such a chart  $(U, \varphi)$  of  $M$  is said to be adapted to  $S$ . Then, we say that  $S$  is a *regular submanifold* of  $M$  of dimension  $k$  if for all  $p \in S$ , there exists a chart  $(U, \varphi)$  of  $M$  with  $p \in U$  such that  $(U, \varphi)$  is adapted to  $S$ . Therefore, if  $(U, \varphi)$  is adapted to  $S$ , then  $\varphi(U \cap S)$  consists of only those points that look like  $(r^1, \dots, r^k, 0, \dots, 0) \in \mathbb{R}^n$ , and hence  $\varphi(U \cap S)$  is an open subset of  $\mathbb{R}^k$ . If we define  $\varphi_S : U \cap S \rightarrow \mathbb{R}^k$  as  $\varphi_S(p) = (x^1(p), \dots, x^k(p))$ , then  $\varphi_S(U \cap S) = \varphi(U \cap S) \cap \mathbb{R}^k$ .

**Theorem 1.43.** Consider the maximal atlas  $\mathcal{U} = \{(U, \varphi)\}$  of  $M$ , and let  $\mathcal{A} = \{(U \cap S, \varphi_S) \mid (U, \varphi) \in \mathcal{U}\}$ , where  $(U, \varphi)$  is adapted to  $S$  and  $\bigcup \{U \cap S \mid (U, \varphi) \in \mathcal{U}\} = S$ . Then  $\mathcal{A}$  is a smooth atlas on  $S$ , and hence  $S$  is a smooth manifold of dimension  $k$ .

The *codimension* of a regular submanifold  $S$  of  $M$  is defined as  $\text{codim}_M S = \dim(M) - \dim(S) = n - k$ .

Let  $f : M \rightarrow \mathbb{R}$  be a smooth function on a smooth manifold  $M$  of dimension  $n$ . Also let  $0 \in \mathbb{R}$  be a regular value of  $f$  with  $f^{-1}(0) \neq \emptyset$ . Then,  $f^{-1}(0)$  is a regular submanifold of  $M$  of dimension  $n - 1$ , or codimension 1. Note that 0 here is a red herring; we can replace 0 with any regular value  $c \in \mathbb{R}$  of  $f$  with  $f^{-1}(c) \neq \emptyset$ , and the same result holds.

For a proof of this, let  $p \in f^{-1}(0) \neq \emptyset$  so that  $f(p) = 0$ . Since 0 is a regular value of  $f$ , we have that  $f_{*,p} : T_p M \rightarrow T_0 \mathbb{R}$  is surjective.  $T_p M$  has a basis  $\{\frac{\partial}{\partial x^1}|_p, \dots, \frac{\partial}{\partial x^n}|_p\}$ , and  $T_0 \mathbb{R}$  has a basis  $\{\frac{d}{dt}|_0\}$ . The  $1 \times n$  Jacobian matrix of  $f$  at  $p$  is given by

$$(df)_p = \left( \frac{\partial f}{\partial x^1}(p), \dots, \frac{\partial f}{\partial x^n}(p) \right). \quad (1.77)$$

Since  $f_{*,p} : T_p M \rightarrow T_0 \mathbb{R}$  is surjective, we have that  $\text{rank}(f_{*,p}) = 1$ , and hence  $(df)_p \neq 0$ ; at least coordinate function  $x^i$  has  $\frac{\partial f}{\partial x^i}(p) \neq 0$ . Without the loss of generality, we may assume that  $\frac{\partial f}{\partial x^1}(p) \neq 0$ . Then, by the inverse function theorem, there exists an open neighbourhood  $V$  of  $p$  such that  $f : V \rightarrow f(V) \subseteq \mathbb{R}$  is a diffeomorphism. Let  $(U, \varphi) = (U, x^1, \dots, x^n)$  be a chart on  $M$  with  $p \in U \subseteq V$ . Then, we have

$$U \cap f^{-1}(0) = \{(x^1, \dots, x^n) \in U \mid x^1 = 0\}, \quad (1.78)$$

which shows that  $(U, \varphi)$  is adapted to  $f^{-1}(0)$ . Since  $p \in f^{-1}(0)$  was arbitrary, we have that  $f^{-1}(0)$  is a regular submanifold of  $M$  of dimension  $n - 1$ , or codimension 1. In general, this result generalizes to the following theorem.

**Theorem 1.44.** Let  $F : N \rightarrow M$  be a smooth map between smooth manifolds of dimension  $n$  and  $m$ , respectively, with  $n \geq m$ . Let  $q \in M$  be a regular value of  $F$  with  $F^{-1}(q) \neq \emptyset$ . Then,  $F^{-1}(q)$  is a regular submanifold of  $N$  of dimension  $n - m$ , or codimension  $m$ .

*Proof.* Let  $p \in F^{-1}(q)$ ; then  $F_{*,p} : T_p N \rightarrow T_q M$  is surjective. Let  $(U, \varphi) = (U, x^1, \dots, x^n)$  be a chart in  $N$  such that  $\varphi(p) = 0 \in \mathbb{R}^n$ , and let  $(V, \psi) = (V, y^1, \dots, y^m)$  be a chart in  $M$  such that  $\psi(q) = 0 \in \mathbb{R}^m$ .

We may assume  $F(U) \subseteq V$ . The matrix of  $F_{*,p}$  with respect to the bases  $\{\frac{\partial}{\partial x^1}|_p, \dots, \frac{\partial}{\partial x^n}|_p\}$  and  $\{\frac{\partial}{\partial y^1}|_q, \dots, \frac{\partial}{\partial y^m}|_q\}$  is given by  $\left(\frac{\partial F^i}{\partial x^j}(p)\right)_{1 \leq i \leq m, 1 \leq j \leq n}$ ; this has rank  $m$  since  $F_{*,p}$  is surjective. Without the loss of generality, we may assume that the  $m \times m$  submatrix  $\left(\frac{\partial F^i}{\partial x^j}(p)\right)_{1 \leq i, j \leq m}$  is invertible. Then, by the inverse function theorem, there exists an open neighbourhood  $W$  of  $p$  such that  $F : W \rightarrow F(W)$  is a diffeomorphism. Let  $(U, \varphi) = (U, x^1, \dots, x^n)$  be a chart on  $N$  with  $p \in U \subseteq W$ . Then, we have

$$U \cap F^{-1}(q) = \{(x^1, \dots, x^n) \in U \mid x^1 = x^2 = \dots = x^m = 0\}, \quad (1.79)$$

which shows that  $(U, \varphi)$  is adapted to  $F^{-1}(q)$ . Since  $p \in F^{-1}(q)$  was arbitrary, we have that  $F^{-1}(q)$  is a regular submanifold of  $N$  of dimension  $n - m$ , or codimension  $m$ . ■

*August 19th.*

**Example 1.45.** Consider the map  $f : \mathbb{R}^3 \rightarrow \mathbb{R}$  defined as  $f(x, y, z) = x^2 + y^2 + z^2$ . Then,  $S^2 = f^{-1}(1) = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$ . We have  $f_{*,p} : T_p \mathbb{R}^3 \rightarrow T_1 \mathbb{R} = \mathbb{R}$  given by  $f_{*,p}(\frac{\partial}{\partial x^i}|_p) = \sum_{j=1}^3 \frac{\partial f}{\partial x^j}(p) \frac{\partial}{\partial y}|_1$ , where  $y$  is the coordinate function on  $\mathbb{R}$ . Then, the Jacobian matrix of  $f$  at  $p$  is given by  $(df)_p = \left(\frac{\partial f}{\partial x}(p), \frac{\partial f}{\partial y}(p), \frac{\partial f}{\partial z}(p)\right) = (2x, 2y, 2z)$ . The origin is the only critical point of  $f$  and is excluded from  $S^2$ . Thus,  $1 \in \mathbb{R}$  is a regular value of  $f$ , and hence  $S^2 = f^{-1}(1)$  is a regular submanifold of  $\mathbb{R}^3$  of dimension  $3 - 1 = 2$ .

**Example 1.46.** Consider the determinant map  $\det : \mathbb{R}^{n^2} \rightarrow \mathbb{R}$ , interpreting a point  $p \in \mathbb{R}^{n^2}$  as an  $n \times n$  matrix  $A \in M_n(\mathbb{R})$ . Look at  $SL_n(\mathbb{R}) = \det^{-1}(1) = \{A \in M_n(\mathbb{R}) \mid \det(A) = 1\}$ . We have  $\det_{*,A} : T_A \mathbb{R}^{n^2} \rightarrow T_1 \mathbb{R} = \mathbb{R}$ , where the bases are  $\{\frac{\partial}{\partial x_{ij}}\}_{1 \leq i, j \leq n}$  and  $\{\frac{d}{dt}|_1\}$ , respectively. Note that  $\frac{\partial \det}{\partial x_{ij}}(A)$  is the  $(i, j)$ -th cofactor of  $A$ ; not all of these cofactors can be zero since otherwise we would have  $\det(A) = 0$  but  $A \in \det^{-1}(1)$ . Thus,  $1 \in \mathbb{R}$  is a regular value of  $\det$ , and hence  $SL_n(\mathbb{R}) = \det^{-1}(1)$  is a regular submanifold of  $\mathbb{R}^{n^2}$  of dimension  $n^2 - 1$ .

The following is a corollary of the inverse function theorem; it may be generalized to the case of smooth maps between smooth manifolds.

**Corollary 1.47.** Let  $p \in \mathbb{R}^n$  have an open neighbourhood  $U \subseteq \mathbb{R}^n$ , and let  $f : U \rightarrow \mathbb{R}^m$  be a smooth map with  $f = (f_1, \dots, f_m)$ . Suppose  $f$  has rank  $k$  in a neighbourhood of  $p$  in  $U$ ; that is,  $\left(\frac{\partial f_i}{\partial x_j}(q)\right)$  has rank  $k$  for all  $q$  in this possibly smaller neighbourhood of  $p$ . Then there exists a diffeomorphism  $G : V \rightarrow G(V)$ , with  $G(V)$  open in  $\mathbb{R}^n$  and  $p \in V$  open in  $U$ , and a diffeomorphism  $F : W \rightarrow F(W)$ , with  $F(W)$  open in  $\mathbb{R}^m$  and  $f(p) \in W$  open in  $\mathbb{R}^m$ , such that  $f(V) \subseteq W$ , and  $(FfG^{-1})(x^1, \dots, x^n) = (x^1, \dots, x^k, 0, \dots, 0)$  for all  $(x^1, \dots, x^n) \in G(V)$ .

*Proof.* Let  $p \in \mathbb{R}^n$  have an open neighbourhood  $U \subseteq \mathbb{R}^n$ , and let  $f : U \rightarrow \mathbb{R}^m$  be a smooth map with  $f = (f_1, \dots, f_m)$ . Suppose  $f$  has rank  $k$  in a neighbourhood of  $p$  in  $U$ . Then, without the loss of generality, we may assume that the  $k \times k$  submatrix  $\left(\frac{\partial f_i}{\partial x_j}(q)\right)_{1 \leq i, j \leq k}$  is invertible for all  $q$  in this possibly smaller neighbourhood of  $p$ . Define the smooth map  $F : U \rightarrow \mathbb{R}^n$  as  $F(x^1, \dots, x^n) = (f_1(x^1, \dots, x^n), \dots, f_k(x^1, \dots, x^n), x^{k+1}, \dots, x^n)$ . Then, the Jacobian matrix of  $F$  at  $q \in U$  is given by

$$(dF)_q = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(q) & \cdots & \frac{\partial f_1}{\partial x_k}(q) & \frac{\partial f_1}{\partial x_{k+1}}(q) & \cdots & \frac{\partial f_1}{\partial x_n}(q) \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_k}{\partial x_1}(q) & \cdots & \frac{\partial f_k}{\partial x_k}(q) & \frac{\partial f_k}{\partial x_{k+1}}(q) & \cdots & \frac{\partial f_k}{\partial x_n}(q) \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \quad (1.80)$$

Thus, by the inverse function theorem, there exists an open neighbourhood  $V$  of  $p$  with  $V \subseteq U$  such that  $F : V \rightarrow F(V)$  is a diffeomorphism. Consider the map  $F^{-1} : F(V) \rightarrow V$ ; a point  $(x^1, \dots, x^n) \in F(V)$  is

mapped to  $(x^1, \dots, x^k, f^{k+1}, \dots, f^m)$ . The Jacobian matrix of  $F^{-1}$  at  $(x^1, \dots, x^n) \in F(V)$  is given by

$$(dF^{-1})_{(x^1, \dots, x^n)} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ * & * & * & \frac{\partial f^{k+1}}{\partial x^{k+1}} & \dots & \frac{\partial f^{k+1}}{\partial x^n} \\ * & * & * & \vdots & \ddots & \vdots \\ * & * & * & \frac{\partial f^m}{\partial x^{k+1}} & \dots & \frac{\partial f^m}{\partial x^n} \end{pmatrix}. \quad (1.81)$$

We know this matrix to have rank  $k$  by hypothesis. Thus, the  $(m-k) \times (n-k)$  submatrix  $\left(\frac{\partial f^i}{\partial x^j}(q)\right)_{k+1 \leq i \leq m, k+1 \leq j \leq n}$  must have rank 0; that is,  $\frac{\partial f^i}{\partial x^j}(q) = 0$  for all  $k+1 \leq i \leq m$  and  $k+1 \leq j \leq n$ . The functions  $f^{k+1}, \dots, f^m$  are then functions of only  $x^1, \dots, x^k$ . Define the smooth map  $G : f(V) \rightarrow \mathbb{R}^m$  as  $(y^1, \dots, y^m) = (y^1, \dots, y^k, y^{k+1} - f^{k+1}, \dots, y^m - f^m)$ . Then, the Jacobian matrix of  $G$  at  $(y^1, \dots, y^m) \in f(V)$  is given by

$$(dG)_{(y^1, \dots, y^m)} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ -\frac{\partial f^{k+1}}{\partial y^1} & \dots & -\frac{\partial f^{k+1}}{\partial y^k} & 1 & 0 & 0 \\ \vdots & \ddots & \vdots & 0 & 1 & 0 \\ -\frac{\partial f^m}{\partial y^1} & \dots & -\frac{\partial f^m}{\partial y^k} & 0 & 0 & 1 \end{pmatrix}. \quad (1.82)$$

This matrix is of full rank, so  $G$  is a diffeomorphism from  $f(V)$  to  $G(f(V))$ . Then, we have  $(GF)(x^1, \dots, x^n) = (f_1(x^1, \dots, x^n), \dots, f_k(x^1, \dots, x^n), 0, \dots, 0)$  for all  $(x^1, \dots, x^n) \in V$ . Thus, we have found diffeomorphisms  $G$  and  $F$  such that  $(GF)(x^1, \dots, x^n) = (x^1, \dots, x^k, 0, \dots, 0)$  for all  $(x^1, \dots, x^n) \in G(V)$ . ■

The generalization is as follows. Let  $f : N \rightarrow M$  be a smooth map between smooth manifolds of dimension  $n$  and  $m$ , respectively. Let  $p \in N$  be a point such that  $f$  has a rank  $k$  in a neighbourhood of  $p$  in  $N$ . Then there exists a chart  $(U, \varphi)$  centred at  $p$  in  $N$  and a chart  $(V, \psi)$  centred at  $f(p)$  in  $M$  such that  $f(U) \subseteq V$ , and  $\psi \circ f \circ \varphi^{-1}(x^1, \dots, x^n) = (x^1, \dots, x^k, 0, \dots, 0)$  for all  $(x^1, \dots, x^n) \in \varphi(U)$ .

**Theorem 1.48.** *Let  $f : N \rightarrow M$  be a smooth map between smooth manifolds of dimension  $n$  and  $m$ , respectively. Let  $m \in M$  be such that  $f$  has a rank  $k$  in a neighbourhood of  $f^{-1}(m)$  in  $N$ . Then  $f^{-1}(m)$  is a regular submanifold of  $N$  of dimension  $n - k$ .*

**Example 1.49.** Look at the map  $f : GL_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R})$  define by  $f(A) = A^t A$ . Then,  $O_n(\mathbb{R}) = f^{-1}(I_n) = \{A \in GL_n(\mathbb{R}) \mid A^t A = I_n\}$ . For now, the dimension of  $O_n(\mathbb{R})$  will remain unknown, but we can still show that it is a regular submanifold of  $GL_n(\mathbb{R})$ . Let  $C \in GL_n(\mathbb{R})$ . Then  $f(AC) = C^t f(A)C$ . Note that conjugation by  $C$  is a diffeomorphism from  $GL_n(\mathbb{R})$  to itself, and hence  $f$  has the same rank at  $A$  and  $AC$ . We shall show that the ranks of  $f_{*,A}$  and  $f_{*,B}$  are the same for all  $A, B \in GL_n(\mathbb{R})$ . We can break down  $B = AC$  for some  $C \in GL_n(\mathbb{R})$ . Then  $f \circ r_C = l_{C^t} \circ r_C \circ f$ , where  $r_C : GL_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R})$  is right multiplication by  $C$  and  $l_{C^t} : GL_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R})$  is left multiplication by  $C^t$ . Then, we have

$$f_{*,AC}(r_C)_{*,A} = (l_{C^t})_{*,A^t AC}(r_C)_{*,A^t A} f_{*,A}. \quad (1.83)$$



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